

Contents

Table of Contents

Contents	1
Table of Contents	1
List of Acronyms	1
List of Figures	1
Introduction	2
An overview of the literature	3
Motivations & Work Objectives	6
Contents of the thesis	7
CHAPTER 1: Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers	9
1.1 Modelling of porous media	9
1.2 Modelling of the wave propagation in homogeneous layers	13
1.3 Methods for modelling metaporous structures	17
CHAPTER 2: A general spectral collocation method for computing the dispersion relations of guided acoustic waves in multilayer dissipative structures	20
2.1 Abstract	21
2.2 Introduction	21
2.3 Solving the dispersion relations for multilayer structures	22
2.4 Application to a two-layer poroelastic-elastic structure	26
2.5 Conclusion	33
2.6 Appendices	34
Appendices	40
A Benchmark of metaporous configurations	41
B Implementing SAFE to compute the dispersion relation of Lamb modes	41
C Details on the root-finding algorithm	44
D Overview of activities and scientific works during the thesis	46
E Extended abstract in french	47

List of Acronyms

BEM Boundary Elements Method.	PEM Poroelastic material.
EPWE Extended Plane Wave Expansion.	PML Perfectly Matched Layer.
FEM Finite Elements Method.	PWE Plane Wave Expansion.
GEP Generalized Eigenvalue Problem.	SAFE Semi-analytical Finite Elements.
GM Global Method.	SCM Spectral Collocation Method.
JCA Johnson-Champoux-Allard.	SRR Split-Ring Resonator.
MST Multiple Scattering Theory.	TMM Transfer Matrix Method.

List of Figures

I.1	Periodic cells in 1, 2 and 3D. Image taken from Ref. ²⁷	3
I.2	Example of metaporous configurations	5
I.3	Diagram presenting a general layout for the configurations that will be studied in each chapter	7
1.1	Porous material structure	10
1.2	Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.	13
1.3	Diagram with results of the reflection and transmission coefficients for equivalent fluid and poroelastic layers	16
1.4	Discretization for each method	17
2.1	a) Sketch of the geometry of the multilayer system and b) zoom on the n -th layer.	22
2.2	a) Sketch of the two-layer structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. Blue dots represents the results as calculated with the SCM and black open markers represents those as calculated with the Müller method. Bulk wavenumbers in the poroelastic and elastic materials are highlighted in yellow and dark red respectively.	29
2.3	a) and b) represent the real and imaginary parts of the dispersion relation of the two-layer structure. Two branches are highlighted in yellow and green, as well as two specific solutions, k_1^- (blue) and k_1^+ (red), symmetric with respect to the $\text{Re}(k_1) = 0$ axis. Displacement fields in the x_1 c) and x_2 direction d) associated with the k_1^+ and k_1^- solutions displayed. Total Poynting e) vector and fluid Poynting vector f) with the length of the arrow indicating the amplitude and angle indicating the direction of the vector field. g) Group and energy transport velocities for the yellow branch. h) Energy transport velocity for the green branch. The gray area represents the frequency range over which this velocity is negative.	32
2.4	a) Experimental set-up used for the measurement of the normal displacement field of the two-layer structure, with a close-up of the sample excitation. b) Real and c) imaginary parts of the dispersion relations recovered experimentally from the SLaTCoW method k_i (black crosses) and calculated from the SCM k_i (colored lines and black-to-grey dots) and the bulk wavenumbers in the poroelastic material (magenta lines).	33
2.5	a) Sketch of the single layer poroelastic structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. The branches of the symmetric and antisymmetric modes are represented by blue and red curves respectively. The dispersion relation of the two-layer elastic-poroelastic system studied in Section 2.4 are reminded with black lines.	35
2.6	Description of the procedure of the SLaTCoW method: a) Spatial Laplace transform from the experimental data, b) Spatial Laplace transform of the reconstructed wave fields. Real c) and imaginary d) parts of the experimentally retrieved wavenumbers (crosses) compared to the SCM (black-to-grey dots). The frequency at which the Laplace transforms are represented corresponds to the magenta dashed line in the graph.	38
2.7	a) Dispersion relation for the Lamb modes with the root-finding results, alongside the spectral collocation and semi-analytical finite elements solutions. b) Convergence curve of the 2 methods at a given frequency $f = 4$ kHz. The multiple lines depicts the convergence of each wavenumber.	43
2.8	iteration muller	44
2.9	Description of the routine	45

Introduction

Noise is a major issue in our modern societies, ranging from everyday life inconveniences, work comfort problems to an actual pollution problem in severely affected areas. Sources of noise can be diverse; in urban environments, concerns are mostly on traffic and population noise. From industrial origins, noise can have various consequences, causing not only risks to the workers and the neighbouring populations, but also structural hazard to installations. Generally, motorized transport is a major issue in all environments, including aircraft, public transportation, roads, commercial vessel traffic, etc.

The European Environment Agency estimated in 2021 that 1 in 5 European resident are exposed to unhealthy levels of noise, its main cause being traffic. These noise exposures can have a strong impact on health¹ favoring impaired circadian rhythm, oxidative stress and cardiovascular dysfunction², as well as psychological consequences like anxiety and depression³. Moreover, anthropic noise pollution also affects biodiversity at large, causing stress, behaviour changes, and masking sensory perception⁴. Marine traffic was shown to generate stress on marine mammals and modify their behaviour by provoking area avoidance and vocalization disruption⁵.

Although sound intensity can be measured objectively, the impact of noise on individuals is in part shaped by social and cultural context. Studies show that expected behaviours norms and customs strongly influence reported annoyance⁶, with comparative surveys revealing clear cross-cultural differences. More recent work has linked collectivistic cultural norms to systematically higher everyday noise exposure as an example^{7,8}. Through these examples, we can conclude that noise is a very broad topic that raises complex questions in various scientific disciplines, regarding its impact on human life and biodiversity at large. In this work, we want to focus on noise mitigation using engineered structures in acoustics.

Most noise sources have emerged from the modernisation of societies through various innovations; with them come with the need to mitigate their noise impact up to our current capabilities, which is able to evolve thanks to research on noise barriers and acoustic insulation systems. Various strategies have been applied throughout the ages. Dating back to 4th century BC, acoustic jars were used to improve the room acoustics in Greek and Roman theatres as well as in the Middle Ages in Western Europe in churches⁹. These resonators are thought to be capable of reducing reverberation time in buildings, covering a frequency range matching that of human spoken voice. As another example, a famous roman amphitheater was shown to have a seat row disposition able to improve intelligibility of speeches from the stage¹⁰.

Contemporary to these antique devices, porous and fibrous materials have been traditionally used to absorb sound in the last century¹¹⁻¹⁶. Largely used in the industry, they are efficient at frequencies starting close to the quarter wavelength resonance frequency of the layer and have been extensively analyzed in the literature¹⁷⁻²¹. They can be found in natural occurrences or engineered and cover very large use cases of acoustic insulation such as industry, transport, aerospace, concert halls and many others.

An overview of the literature

A brief history of Acoustic Metamaterials

In more recent years, the emergence of more complex engineered structures has disrupted the landscape of acoustic absorbers. Metamaterials are composite structures designed to exhibit physical effective properties that go beyond the capability of the sole material that composes the bulk. They can be created by a lot of various means, including engineered microstructure, a gradual evolution of the material properties, or repeating patterns or scatterers implemented in one or several directions in space and embedded in a host material as shown for various dimensions in Fig. I.1. Compared to homogeneous layers, they generally lead to more complex structures, and are able to achieve enhanced effects. Usually allowing for a tunable response, these structures offer great flexibility and open new perspectives in material design. This field has been initiated by Veselago, who has theoretically described the electrodynamics of materials having a negative refraction index (occurring in the case of a doubly-negative permittivity and permeability), now known as the Veselago lens in electromagnetic metamaterials²². These discoveries have subsequently sparked interest in various disciplines, such as photonics, optics²³ and in the case that concerns us, acoustics. Metamaterials can be used to design antennas and absorber for electromagnetic waves and seismic protection devices among others. In acoustic, they allow for various applications such as absorption, filtering, diffusion or cloaking.

Seminal works have been done on the study of the use of lattice arrangements or crystals for wave control applications. Photonic crystals are used to achieve the control of light and electromagnetic waves²⁴. Similarly in acoustics, phononic crystals have been developed for the control of elastic and acoustic waves as the 2D (or 3D) periodic distribution of scatterers in a solid host medium²⁵ as depicted on Fig. I.1. In the case where the host medium is constituted by a fluid, these structures are called sonic crystals²⁶. These structures are often studied through their band structure (also called dispersion diagram/relation when applied to the study of non-periodic geometries), which will be discussed in extensive details in Section [ref this in chap1](#). In the remaining of the literature overview, focus will be centered around metamaterials that involve periodicity in only one direction in most cases. These acoustic metamaterials can be divided into two main categories: the first concerns the so-called *reflection problem* where the case study is that of an incoming wave on some sample or panel which has a rigid-backing at the end. There, the consideration is focused on mitigating the reflected wave and achieving important acoustic absorption. The other case is that of a *transmission problem* where the incoming wave is able to transmit through the sample. In this case and depending on the application, either or both reflection and transmission are sought to be mitigated. In each of these cases respectively, the structures can be called metasurfaces and interfaces.

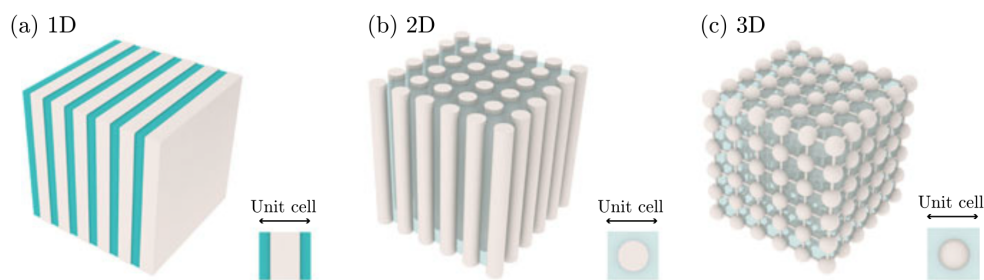


Fig. I.1: Periodic cells in 1, 2 and 3D. Image taken from Ref.²⁷

Structured materials that tackle the reflection problem can involve many various strategies. The combination of a perforated plate and coiled-up channels is able to achieve large broadband absorption underwater²⁸. Micro-perforated plate can also be used to achieve interesting absorption

properties for airborne acoustics²⁹. Materials structured from resonant building blocks with resonance frequencies in the sub-wavelength regime i.e., for frequencies which wavelength is much bigger than the typical size of the material, have also been studied^{30–32}. Large width elastic bandgaps have been achieved using locally resonant cells in the design of a sonic crystal³³. These resonances introduce strong dispersion of the waves and the properties of the system are thus dramatically affected. Together with the unavoidable losses, the resonant behaviour of metamaterials make them perfect candidates to design efficient absorbers at low frequencies with reduced dimensions^{34,35}. Balancing the dissipation and leakage of each resonance yields the recently described *critical coupling*. In this case, the reflection drops to zero³⁶. Perfect absorbers can thus be designed for reflection problems for both monochromatic waves^{34,37} and broad range of frequencies^{37–41}. This can be achieved by tuning the structure using the scattering matrix zeros plotted in the complex frequency plane; critical coupling would be achieved when the scattering matrix zero has a vanishing imaginary frequency. Similarly to these broadband absorbers, Yang *et al.* have realized the backing of a porous layer using Fabry-Pérot channels to design an efficient structure⁴².

One of the initial interests of the acoustic metamaterials community was the possibility of producing deeps in the transmission spectra at very low frequencies. This has been widely developed in the literature, a small subset of these works include Refs. ^{43–46}. These configurations are based on resonant elements and display a zero transmission in the lossless case at their resonance frequency. In reality, losses cannot be avoided and a small amount of the energy is always transmitted. Membrane-type acoustic metamaterials^{47–49}, or acoustic metamaterials made by Helmholtz resonators^{50–52} have been used to develop systems with high values of transmission loss for audible acoustics. Considering the vibroacoustic coupling allows for more precise depiction of real-life applications. Works on periodic soft inclusion in a soft solid matrix was done by Aberkane *et al.* showing improved transmission loss affected by the presence of soft inclusions in the solid layer. In this case, the presence of losses was able to improve the transmission loss at low frequencies. Also accounting for vibroacoustic coupling, attached resonators to walls have been also used to reduce the transmission loss of panels^{53,54}.

For the case of mirror-symmetric structures, it is worth noting here that if only a monopolar resonator (Helmholtz) or a dipolar resonator (membrane) is used to absorb waves, then only 50% of energy can be absorbed as shown in several articles^{55,56}. However, using a material made of slits loaded by an array of identical resonators creates an accumulation point close to the resonance frequency of the resonators allowing the accumulation of the resonances of the slit, thus placing together symmetric and antisymmetric resonances^{47,50,57}, although requiring for a compromise between absorption and thickness. A more complex approach consists in designing degenerate resonators. By using combinations of membrane and cavity resonators^{47,58} and with slits loaded with Helmholtz resonators⁵⁶, quasi-total absorption can be reached. In these systems, each resonator manages half of the problem, absorbing each one half of the energy and allowing a perfect absorption peak: all the energy that impinges on the system is dissipated within the system, i.e., the energy is neither reflected nor transmitted. The other alternative to obtain perfect absorption consists in breaking the symmetry of the system. These systems can achieve unidirectional perfect absorption, i.e., perfect absorption can be obtained from one of the two incidence sides⁵⁵. The principle of these systems consists in using two resonators, one that is used to reduce as much as possible the transmission and the other that impedance matches the system from one side^{34,37,49}. This concept has been exploited to develop broadband perfect absorbers using a cascade of the previous process giving rise to a rainbow absorber³⁸.

From porous layers to metaporous surfaces

As mentioned in the beginning of the introduction, porous materials have been traditionally widely used for their acoustic insulation performances. These materials are lossy: an airborne acoustic wave propagating through a porous layer is dissipated as it propagates in the bulk in the form of

viscous friction due to the drag between the air and the pore walls and in the form of heat due to the small ducts and the existence of a thermal boundary layer. These losses mechanisms are strongly dependant on the nature of the material, as they are mainly affect by the geometry of the microstructure. Common models usually account for homogenized effective parameters to characterize these materials as will be discussed in a further section. Since wave velocity in porous media tends to be smaller than that of air, allowing to lower slightly the frequency of the quarter-wavelength resonance, which corresponds due the frequency where such materials usually start to have an efficient acoustic absorption. Section 1.1 is dedicated to presenting the various models used for porous and poroelastic media modelling. Their absorption efficiency depends thus on their inherent losses but also on the impedance mismatch between the layer and the medium from which the impinging acoustic wave is originated (i.e usually air). The acoustic impedance mismatch leads to reflection at the interface, while the energy transmitted through the interface in the porous layer gets attenuated. In real-life applications, these materials are not closed systems and there is a need to account for their coupling irl to surrounding domains into which part of the acoustic energy is susceptible to be radiate back into the half-space.

In order to increase the capabilities of acoustic absorption of porous materials, some works can involve the modification of the core structure of porous and poroelastic materials by tuning of the microstructure, introducing a gradient of the properties, or by using anisotropic materials. Recently, with the increase of new fabrication techniques based on 3D printing, the structure of the porous materials can be controlled and tuned to design materials with better absorption capabilities at low frequencies, based on gradient properties^{59,60}, or on anisotropic materials^{61–64}. Recently, resonators filled with 3D printed porous materials have been developed to design perfect and broadband absorption⁶⁵. Other configurations have been studied, among them the periodic partitioning of the metaporous layer⁶⁶ or a labyrinthine channel embedded in a porous host⁶⁷, based on the idea of coiling up the space of the channel as was done for non-porous metamaterial designs. Recently, a hierarchical multilayer metaporous structure was proposed using an optimization procedure, giving rise to gradient and periodic designs⁶⁸.

Some studies involving porous layers embedding periodic arrays of inclusions were done during the last 15 years^{69–72}. These inclusions can have either resonances or not and they allow for an increased dissipation in the viscous regime. These structures are refered to as metaporous surfaces when used in reflection problems. The first works involving metaporous surfaces or interfaces were developed for transmission problems^{73,74}. In case of more than one inclusion per spatial period, several absorption peaks can be obtained for a particular arrangement along the porous thickness⁷⁰. Nevertheless, this rapidly leads to a thick structure. Near the inclusion resonance frequency, chosen to be in the viscous regime, an absorption peak occurs⁷⁵. The resonance frequency of the inclusions depends on the orientation of the resonators relative to the rigid boundary^{71,75}. Recently, 3D printed porous materials with embedded split-ring resonators have been modeled⁶³, designed and experimentally characterized showing perfect absorption peaks. More complex spiral-shaped slits have been embedded in a porous layer and have shown to be able to maintain a high absorption for large incidence angles⁷⁶. Xiong *et al.* have studied a metaporous liner with embedded resonator and shown the existence of an exceptional point when the geometry is tuned properly, leading to enhanced absorption⁷⁷.

Fig. I.2: Example of metaporous configurations

Previous metaporous layers studies have considered a host medium modelled as a rigid-frame porous layer; extensions of these structures to consider poroelastic materials have been done⁷⁸. Interesting properties were exhibited for thin elastic shell inclusions, and particularly in the case of a coupling between the trapped and the first volume mode leading to a high wide band absorption. While this previous study implements a semi-analytical model that makes use of Multiple Scattering

Theory (MST), it is worth noting here that different modelings exist as well, based on a Finite Elements Method (FEM) and MST have been validated for the case of an air-saturated metaporoelastic configuration with a solid inclusion and rubber coatings⁷⁹. This thesis is especially focused on the computation of the dispersion relation of such configurations. Therefore, a dedicated literature review dealing with the tools available to achieve such computation will be given in Chapter 1.

Wave propagation modelling and dispersion analysis

The analysis of wave propagation in periodic and complex media often relies heavily on numerical tools. Two main topics are of interest in this thesis: the scattering of an incident wave by a structure as well as its dispersion relation. The first one is simply the acoustic response of a system, where an incident wave propagates towards a specific geometry and is scattered by the latter. The scattering properties are of utmost importance, since tuning those is the main topic in metamaterial and the goal is often to reach exceptional properties for these, such as perfect absorption of the incident wave by the structure as an example. This effect can occur because a specific mode of the structure is excited. These scattering properties depend on the nature of the material(s) used, on the geometry (as we have seen, periodicity can strongly affect the propagation) but also the modal behaviour of the whole system. In order to understand in more details this behaviour, we must turn to dispersion relations.

Dispersion relations are diagrams that represent the wavenumber in a given direction against the frequency, thus informing on the evolution of the wave velocities in the structure and thus the dispersion within. In the context of metamaterials and periodic structures, they are often referred to as band structure. A theoretical definition of a dispersion relation consists in the spectrum of all solutions of a set of equations given a set of parameters. This shows the possible existing states of a system. Generally the form of the dispersion relation, and the various modes that are uncovered can give very in-depth details about the modal behaviour of any structures. In ultrasonics and non-destructive testing, localizing zero-group velocity (ZGV) points in the dispersion relation is very helpful for a number of uses including defects localization and material characterization⁸⁰.

Scarce literature focuses on the computation of the dispersion relations of dissipative layers in general exists: the dispersion of waves in a single poroelastic layer has been studied by Boeckx *et al.*^{81,82} while an unpublished paper from Kiefer *et al.* took interest in Lamb waves in viscoelastic layer coupled to surrounding half-spaces⁸³. There is an interest in analogously studying the coupling between surrounding media and metaporous surfaces, as part of the acoustic energy of the wave propagating in the surface should radiate back into the domain, leading to more complex loss effects on the structure.

These aspects will be discussed in detail in the following chapter, both from the theoretical perspective of porous and poroelastic media (and their coupling to surrounding media) and in terms of the available numerical tools for their computation.

Motivations & Work Objectives

This thesis is dedicated to the numerical modeling of wave propagation in open-system structures comprised of stacks of layers, homogeneous or with embedded scatterers. As it has been seen in the previous section, these materials combine the dissipative characteristics of porous media with the geometric or structural features typical of acoustic metamaterials, thus offering advanced control over their scattering properties. Understanding and predicting the acoustic behavior of such systems is therefore essential for designing efficient structures. In particular, this work focuses in part on the calculation of the dispersion of waves in these structures, and understand how the propagating and evanescent modes that exist in these configurations are shaping the scattering properties. Thence, the

development of efficient numerical strategies is critical to this study.

The tuning of metamaterials could be achieved in some studies by looking at the zeros of the reflection coefficient in the complex frequency plane, thus highlighting the zeros of the scattering matrix. However, the wavenumber might be able to give out additional informations regarding the acoustic behaviour of metaporous layers. Band structure gives useful informations concerning sonic and phononic crystals. However, the dispersion relation of a metaporous layer embedded in an open system has never been addressed; both mechanisms of energy leakage in the surrounding media and dissipations in the porous layer should be investigated. However, this imply a heavy modification of the boundary conditions of the structure. It expected to have significant consequences on the propagation of modes therein. This work will be done on both homogeneous layers as well as periodic structures.

From these observations, we should review existing numerical methods able to compute the acoustic response of metaporous structures and consider ways to obtain a precise and robust method able to predict the scattering properties and acoustic responses of metaporous surface and interfaces. Additionally, we should evaluate methods that have been implemented for the computation of guided waves in layers and stacks of layers coupled to semi-infinite media. Confronting the dispersion relation, with the absorption coefficient should gelp highlighting the modal behaviour responsible for the various absorption peaks obtained by previously studied configurations and also highlight the full modes existing in the layers which might uncover additional phenomena.

Contents of the thesis

This introductory section has set the context as well as an overview of the literature in which this work is situated. We saw that acoustic metamaterials have emerged in the last decades and propose noise mitigation structures that are much more efficient that the traditional foams used for sound insulation. In particular, metaporous surfaces have shown to be promising options achieving one or multiple unitary absorption peaks at subwavelength frequency. Overall, a focus area has been identified towards identifying more precisely the modal behaviour of these configurations. Thus, in this thesis the main interest and final is to succesfully compute the dispersion relations of guided waves in such structures. The manuscript is split into four different chapters as follows,

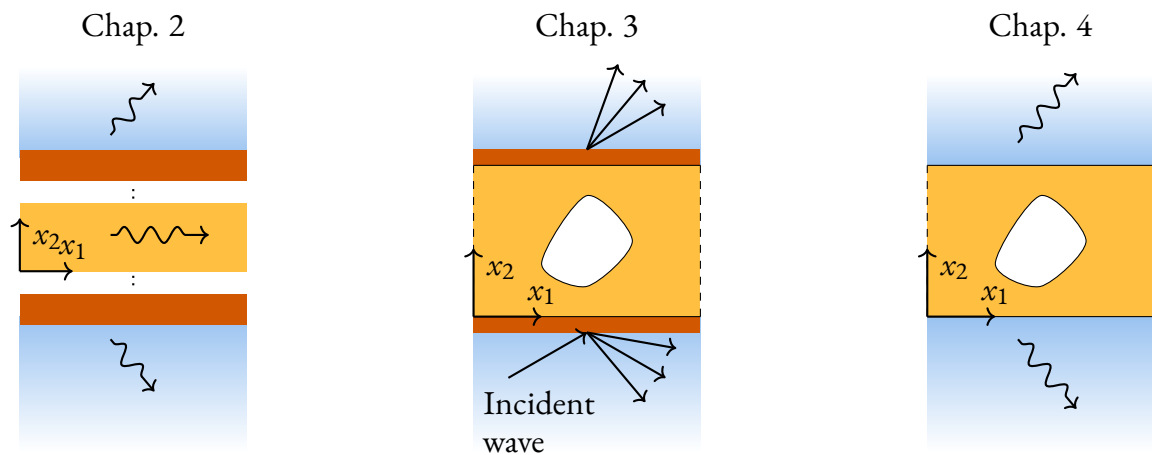


Fig. I.3: Diagram presenting a general layout for the configurations that will be studied in each chapter

- CHAPTER 1 introduces the fundamental concepts necessary for the good understanding of the thesis. This chapter focuses on the scattering of waves on dissipative layers and the guided waves formalism that is used for the calculation of dispersion relations. Additional bibliography is

proposed on these topics as well as a summary of usable numerical methods to compute such problems.

- CHAPTER 2 targets the study of the propagation of guided waves in multilayer arbitrary structures with dissipative layers. A simple numerical method, the Spectral Collocation Method (SCM) is implemented and extra steps allow for the local coupling of surrounding media allows for the computation of the dispersion relation for these configurations. This chapter is constituted from paper^{P1}.
- CHAPTER 3 is focused on several numerical strategies for the modelling of scattering properties of metaporous/metaporoelastic structures. A global method is coupled to a FEM. Results focused on the performance of the proposed method, compared to other similar frameworks is proposed. The convergence of this method is shown for various metaporoelastic surfaces, which geometries were studied in previous papers. This chapter is originally an article referenced as paper.
- In CHAPTER 4, the modal behaviour of metaporous structure is studied. The fem method from the previous chapter is adapted to describe this problem. Results are presented for the case of a rigid inclusion, as well as with embedded Split-Ring Resonator (SRR). Contents for this chapter have been submitted as an article.

Finally, a summary of this work is proposed along with some perspectives on this work, including early results generalizing results from chapter four to include poroelastic hosts.

Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers

Contents

1.1	Modelling of porous media	9
1.1.1	Rigid-frame approximation	10
1.1.2	Elastic motion of the skeleton: Biot theory	11
	Formulations for the motion equations	11
	Propagation in unbounded poroelastic media	12
1.2	Modelling of the wave propagation in homogeneous layers	13
1.2.1	Global Method: plane wave modelling	14
1.2.2	Transfer Matrix Method	15
1.2.3	Guided waves for dispersion problems	15
	Computing the modes with semi-analytical tools	16
	Numerics for mode computations	16
	A bestiary of the wave solutions in dissipative and open waveguides	17
1.2.4	Numerics for the computation of dispersion relations	17
1.3	Methods for modelling metaporous structures	17
1.3.1	Scattering of waves by periodic structures	17
	Media periodization	18
	Overview of the methods	18
1.3.2	Dispersion relations and band structure in periodic lattices	18

This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 | Modelling of porous media

Porous media is a category of materials which encompasses naturally occurring substances such as rocks, soils and biological tissues, and engineered materials like ceramics, metallic foams, rockwool or certain types of polymers. They are considered biphasic, that is constituted of two distinct domains: a pore network Ω_f filled by a saturating fluid embedded in an elastic matrix Ω_s . A diagram of this

microscopic structure is shown in Fig. 1.1. The porosity is defined as the fraction of fluid over the total volume of the material, as $\phi = V(\Omega_f)/V(\Omega)$. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. At microscopic scales, the geometry of the fluid phase thus forms a complex microstructure. Fortunately, at airborne acoustic scales, these materials can often be considered at a meso-/macroscopic level and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that accounts for their complex geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through friction (viscous losses) along the duct/pore network, while thermal exchanges and elastic dissipation take place when the skeleton and the fluid phase are coupled. These losses are thus dependant on the geometry, which parametrized by the porosity, the saturating fluid properties, the pore network size, connectivity and shape, as well as the frame mechanical parameters.

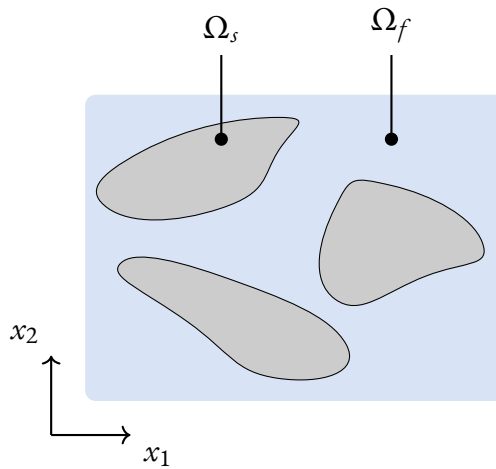


Fig. 1.1: Porous material structure

1.1.1. Rigid-frame approximation

In the so-called rigid-frame approximation, the skeleton of the porous material is considered motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. A harmonic acoustic propagation is assumed with a positive time convention $e^{i\omega t}$ throughout this manuscript (with the exception of Chapter 2). The Helmholtz equation governs the propagation,

$$(\nabla^2 + k_{eq}^2) p(\mathbf{x}) = 0, \quad (1.1)$$

where p is the pressure field and k_{eq} is the wavenumber of the equivalent fluid. In the case where porous or fibrous materials have a porosity close to 1 and the frequency range of interest is limited, the Delany-Bazley model was originally developed in 1970 to propose empirical expressions for the values of the complex wavenumber and characteristic impedances^{11,84} and later improved by Miki⁸⁵. Although limited in accuracy and validity domain, these models have been widely used since they only required the flow resistivity of the porous material. Later on, a more precise semi-phenomenological model additionally accounting for the viscous and inertial dissipative effects was developed from 1987 to 1991. It is the Johnson-Champoux-Allard (JCA) model which models the complex effective density $\rho_{eq}(\omega)$ accounts for the frequency dependant viscous losses and effective bulk modulus of

the material K_{eq} accounting for thermal exchanges in the layer as,

$$\begin{aligned} \rho_{eq}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - i \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda} \right)^2} \right), \\ K_{eq} &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \text{Pr} \omega} \sqrt{1 - i \text{Pr} \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1}. \end{aligned} \quad (1.2)$$

These expressions involve a large number of effective parameters for the porous material; α_∞ is the tortuosity, R_f is the flow resistivity, Λ and Λ' are respectively the viscous and thermal characteristic length. The saturating fluid is characterized in this model by its heat capacity ratio κ (commonly also denoted γ), density ρ_f , kinematic viscosity μ_f , pressure P_0 and Prandtl number $\text{Pr} = c_p \mu_f / k$ with c_p is the specific heat and k the thermal conductivity. We can introduce also the Biot frequency $\omega_c = R_f \phi / (\rho_f \alpha_\infty)$ that defines the transition frequency between the inertial and viscous regimes.

1.1.2. Elastic motion of the skeleton: Biot theory

While the equivalent fluid formalism is able to accurately model porous material in specific sets of conditions, that is when the skeleton can be assumed to be motionless, this assumption can quickly become limited as soon as the said skeleton is subjected to an elastic deformation. This is can be the case when the saturating fluid of the PEM has a high density (e.g. water), when the skeleton is coupled to elastic plates or when the acoustic radiation has a wavelength comparable to the pore scale, although these are just examples. In this situation, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory^{86–89} is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, the various models will be briefly be presented along with the relevant physical quantities. A full derivation would be out of scope and is presented in existing works¹¹.

Before presenting the full Biot model, some quantities should be provided additionally to the equivalent fluid parameters. The frame is assumed to undergo elastic deformations, thus the common mechanical parameters used are the solid density of the frame material ρ_s as well as its apparent density $\rho_1 = (1 - \phi) \rho_s$. The bulk moduli are denoted K_b and K_s , respectively for the frame and the material. The shear modulus is also considered complex, accounting for structural losses as $\tilde{N} = N(1 + i\eta)$ with η the loss factor. Symbols written with a tilde denote complex expressions and therefore account for some sort of loss mechanisms. For the fluid, the apparent density is thus $\rho_2 = \phi \rho_f$ and the inertial interaction density defined by $\rho_{12} = -\phi \rho_f (\alpha_\infty - 1)$. A summary on Biot theory is also proposed in Chap. 2) with the tilde symbols omitted for ease of reading.

Formulations for the motion equations

The Biot poroelastic equations from the original 1956 formulation^{86,88,89} depend on the solid displacement vector $\mathbf{u}_s$ and the fluid displacement $\mathbf{u}_f$,

$$\begin{aligned} \nabla \cdot \boldsymbol{\sigma}_s(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{11} \mathbf{u}_s - \omega^2 \tilde{\rho}_{12} \mathbf{u}_f, \\ \nabla \cdot \boldsymbol{\sigma}_f(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{12} \mathbf{u}_s - \omega^2 \tilde{\rho}_{22} \mathbf{u}_f, \end{aligned} \quad (1.3)$$

where $\boldsymbol{\sigma}^s$ is the solid stress tensor, $\boldsymbol{\sigma}^f$ the fluid stress tensor and the apparent densities of the solid-phase is $\tilde{\rho}_{11}$, that of the fluid is $\tilde{\rho}_{22}$ and the coupling apparent density is $\tilde{\rho}_{12}$. They are complex quantities expression that also account for the contribution of viscous forces and are given as,

$$\tilde{\rho}_{22} = \phi^2 \tilde{\rho}_{eq}, \quad \tilde{\rho}_{12} = \rho_2 - \tilde{\rho}_{22}, \quad \tilde{\rho}_{11} = \rho_1 - \tilde{\rho}_{12}. \quad (1.4)$$

These coupled motion equations are associated to the stress-strain relation,

$$\begin{aligned}\sigma_{s,ij} &= [(\tilde{P} - 2\tilde{N})\nabla \cdot \mathbf{u}_s + \tilde{Q}\nabla \cdot \mathbf{u}_f] \delta_{ij} + 2\tilde{N}\epsilon_{s,ij}, \\ \sigma_{f,ij} &= (\tilde{Q}\nabla \cdot \mathbf{u}^s + \tilde{R}\nabla \cdot \mathbf{u}_f) \delta_{ij} = -\phi p,\end{aligned}\quad (1.5)$$

where $\nabla \cdot \mathbf{u}_s$ and $\nabla \cdot \mathbf{u}^f$ correspond to the dilatation of the solid and fluid phases respectively and ϵ_s is the solid-phase strain tensor. The Biot coefficients are denoted as $\tilde{P}$ for the solid elastic coefficient, $\tilde{Q}$ for the elastic-fluid coupling coefficient, and $\tilde{R}$ for the fluid-phase coefficient. In the context of a PEM saturated by a light fluid, their expressions are

$$\tilde{P} = K_b + \frac{4}{3}N + (1 - \phi)^2 \frac{\tilde{K}_{eq}}{\phi}, \quad \tilde{Q} = \tilde{K}_{eq}(1 - \phi), \quad \tilde{R} = \phi \tilde{K}_{eq}. \quad (1.6)$$

The full coefficients are generalized forms required when the limit $K_b/K_s \rightarrow 0$ no longer holds, as is the case for water-saturated PEM.

A different formulation, especially convenient for finite-elements discretization is known as the $(\mathbf{u}^s, p)$ formulation. First, new parameters in terms of the Biot elastic coefficients and apparent densities can be defined,

$$\hat{A} = \tilde{P} - 2\tilde{N}_s - \frac{\tilde{Q}^2}{\tilde{R}}, \quad \hat{\rho} = \tilde{\rho}_{11} - \frac{\tilde{\rho}_{12}^2}{\tilde{\rho}_{22}}, \quad \beta = \phi \left(\frac{\tilde{\rho}_{12}}{\tilde{\rho}_{22}} - \frac{\tilde{Q}}{\tilde{R}} \right). \quad (1.7)$$

β is another coupling coefficient, also denoted as $\tilde{\gamma}$ in the literature. Using the second row of Eq. (1.3) allows for the expression of the fluid displacement as, $\mathbf{u}^f = \phi/(\tilde{\rho}_{22}\omega^2)\nabla p - \tilde{\rho}_{12}/\tilde{\rho}_{22}\mathbf{u}^s$ and using that in Eq. (1.3) and Eq. (1.5) allow to rewrite the coupled motion equations. Using these, the so-called $(\mathbf{u}_s, p)$ formulation of the Biot theory⁹⁰ is expressed as,

$$\begin{aligned}\nabla \cdot \hat{\sigma}_s + \hat{\rho}\omega^2 \mathbf{u}_s + \beta \nabla p &= 0, \\ \frac{\nabla^2 p}{\rho_{22}\omega^2} - \frac{\beta}{\phi^2} \nabla \cdot \mathbf{u}_s + \frac{1}{R} p &= 0,\end{aligned}\quad (1.8)$$

where $\hat{\sigma}_s$ is the *in vacuo* solid stress tensor,

$$\hat{\sigma}_{s,ij} = \sigma_{s,ij} - \phi \frac{\tilde{Q}}{\tilde{R}} p \delta_{ij} = \hat{A} \nabla \cdot \mathbf{u}_s \delta_{ij} + 2N_s \epsilon_s \quad (1.9)$$

Propagation in unbounded poroelastic media

These various formulations provide several way of modelling equivalently wave propagation in poroelastic media. It appears that PEM supports three waves: a compression φ_p and shear wave ψ_s related to the solid phase as well as an additional compression wave φ_a , related to the fluid phase. In the case of each of these formulations, the fields can be expressed depending on these potentials,

$$\mathbf{u}_s = \nabla \varphi_p + \nabla \varphi_a + \nabla \times \psi_s, \quad \mathbf{u}_f = \mu_p \nabla \varphi_p + \mu_a \nabla \varphi_a + \mu_s \nabla \times \psi_s, \quad (1.10)$$

with μ_p, μ_a, μ_s the solid displacement over the fluid displacement ratio coefficients. The wavenumbers for each of these waves are, following section 6.5 of Allard and Atalla textbook¹¹ and denoted as $\delta_i, i = 1, 2, 3$ in the text,

$$\begin{aligned}k_{p,a} &= \sqrt{\frac{1}{2} \left(\delta_2^2 + \delta_{eq}^2 \pm \delta_t \right)}, \\ k_s &= \omega \sqrt{\frac{\hat{\rho}}{\tilde{N}}},\end{aligned}\quad (1.11)$$

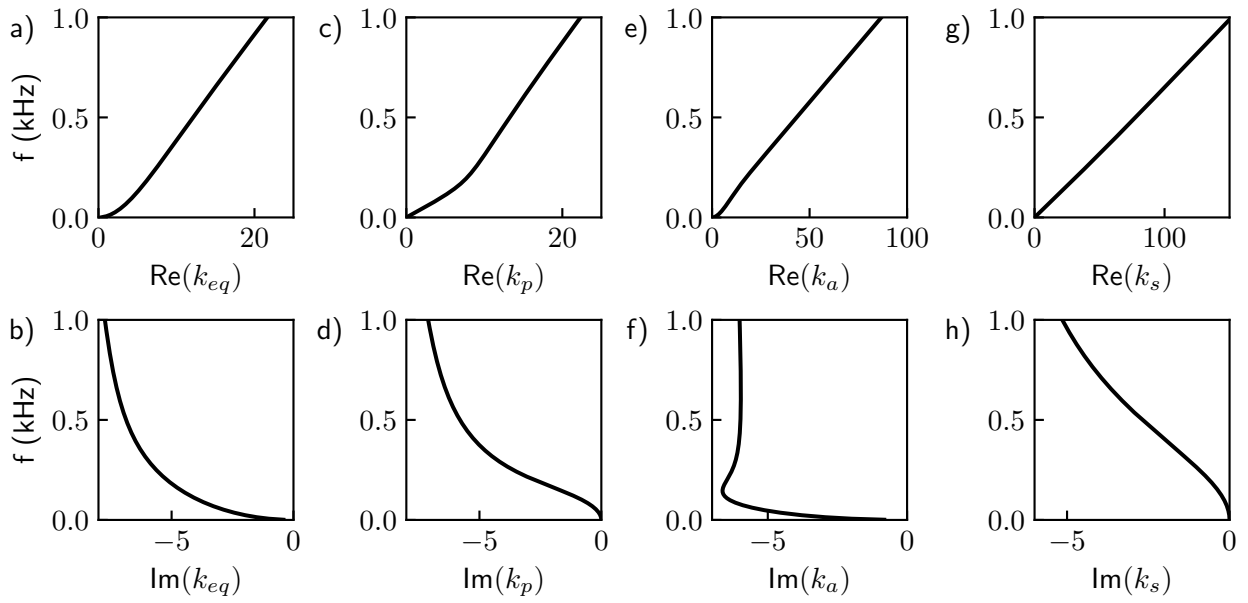


Fig. 1.2: Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.

respectively for the solid compression, fluid compression and shear wave, with,

$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}}}, \quad \delta_1 = \omega \sqrt{\frac{\hat{\rho}}{\hat{P}}}, \quad \delta_2 = \omega \sqrt{\frac{\hat{\rho}_s}{\hat{P}}}, \quad \delta_t = \sqrt{(\delta_2^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2 \delta_1^2}. \quad (1.12)$$

The propagation characteristics, i.e. wavenumbers of the waves propagating in porous and poroelastic media are illustrated in Figure 1.2. As discussed in the previous sections, the wavenumber for the equivalent fluid as well as the three for the poroelastic layer are complex. Their real part account for the propagation of the phase flow, inversly proportional to the wave velocity and their imaginary part accounts for the attenuation that occurs due to the losses inherent to each of these materials. It appears that at low frequencies, the attenuation is very low for all waves. The velocity of the compression and shear waves (observed on the figures by the slope of the real parts of the wavenumbers) is much higher than the waves in the fluid for the equivalent fluid and the compression fluid phase velocity, as they propagate in a soft solid where the velocity is usually much higher than in a fluid.

1.2 Modelling of the wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, which impinges at the surface of a structure and is scattered (reflected and transmitted) through each interface. If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is referred to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they

are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often leads to writing complex system that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave by a homogeneous porous layer is investigated along with the various method available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

1.2.1. Global Method: plane wave modelling

The analytical description of the system amounts finally to a linear system built from the interface conditions, which unknown vector are the amplitudes of each of the existing waves in the system. The forcing right-hand side corresponds to the incident plane. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients.

Introducing a simple problem, that of the scattering of an incident plate by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\mathbf{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1 , there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements.

Assuming a $e^{i\omega t}$ time-dependance, the incident pressure field originated from the upper domain takes the form of,

$$p_{0+} = a_+ e^{-ik_1 x_1} e^{-ik_{2,0}(x_2-h)} + a_- e^{-ik_1 x_1} e^{ik_{2,0}(x_2-h)}. \quad (1.13)$$

The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1 x_1} e^{-ik_{2,eq} x_2} + b_- e^{-ik_1 x_1} e^{ik_{2,eq} x_2}, \quad (1.14)$$

and that in the lower air domain is,

$$p_{0-} = c e^{-ik_1 x_1} e^{ik_{2,0} x_2}, \quad (1.15)$$

with wavenumbers such that they hold for the Snell-Descartes law that states $k_1^2 + k_{2,0}^2 = k_0^2$ and $k_1^2 + k_{2,eq}^2 = k_{eq}^2$. The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -i\omega \rho_i u_2, ::i =$.

$$p^t(h) = p^l(h), \quad u_2^t(h) = u_2^l(h) \quad p^l(0) = p^b(0) \quad \frac{l}{2}(0) = u_2^b(0) \quad (1.16)$$

Imposing the interface conditions leads to the following matrix system,

$$\begin{pmatrix} 1 & e^{ik_{2,eq}h} & 0 & -1 \\ \frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,eq}}{\rho_{eq}} e^{ik_{2,eq}h} & 0 & \frac{k_{2,0}}{\rho_0} \\ e^{ik_{2,eq}h} & 1 & -1 & 0 \\ \frac{k_{2,eq}}{\rho_{eq}} e^{ik_{2,eq}h} & -\frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ a_+ \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_- \\ \frac{k_{2,eq}}{\rho_{eq}} a_- \end{pmatrix}. \quad (1.17)$$

vérifier l'écriture du système. Imposing the wavenumber $k_1 = k_0 \sin \theta$, $k_2 = -k_0 \cos \theta$, with θ the angle of incidence, allows to easily solve the system and obtain all scattered amplitudes b_+ , b_- , a_+ and c . In the case of a rigidly-backed layer, the interface at $x_2 = h$ is replaced by a rigid termination, that is

$u_2(b) = 0$ and thus the terms depending on c vanishes, since p_{0-} does not exist; this reduces the size of the system. In this case, the reflection coefficient R corresponds to the amplitude a_+ .

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in a symmetric configurations are represented by the diagrams in Figs 1.3a and b and results in Figs 1.3c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^\circ$ angle. The parameters for each material corresponds to the one used in Table 2.1 from Chapter ???. The system used to obtain the reflection coefficient for the PEM layer is done in a similar fashion as the equivalent fluid and can be read in the appendices of⁷⁸.

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

Generally, the application of the Global Method allows to write the propagation of plane waves through any stack of layer, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\mathbf{M}(\omega)\mathbf{x} = \mathbf{f}, \quad (1.18)$$

1.2.2. Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson^{thomson1950} and Haskell^{haskell1951} in order to describe elastic wave propagation in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics. This method allows for an easy multiphysics coupling with interface matrices leads to large systems.

The core of the method is in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface. propagate the properties stored in the state vector from one layer to the next. An amount of N layers of the same nature are easy to manipulate since $\mathbf{T}_{\text{global}} = \prod_{n=0}^N \mathbf{T}_n$. This method was deep dived in several textbooks among which we propose Refs.^{11,27}. It allows for the computation of complex system assembly of elements by introducing additional elements.⁹¹. While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes.

$$\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} \cos(k_{eq}h) & iZ_{eq} \sin(k_{eq}h) \\ iZ_{eq}^{-1} \sin(k_{eq}h) & \cos(k_{eq}h) \end{pmatrix} \begin{pmatrix} p(b) \\ u_2(b) \end{pmatrix} \quad (1.19)$$

From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients.

The absorption coefficient for the same two configurations as the last part is presented in Figs 1.3c and d with the markers. It can be seen that the same results are obtained.

1.2.3. Guided waves for dispersion problems

From the previously obtained linear systems in Section 1.2.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, representing the guiding of waves in the layer. This set of equations becomes singular when its determinant vanishes for a given parameter, allowing to identify the modes of the structure. Several formalism can be adopted: solving for frequencies while imposing the wavenumber

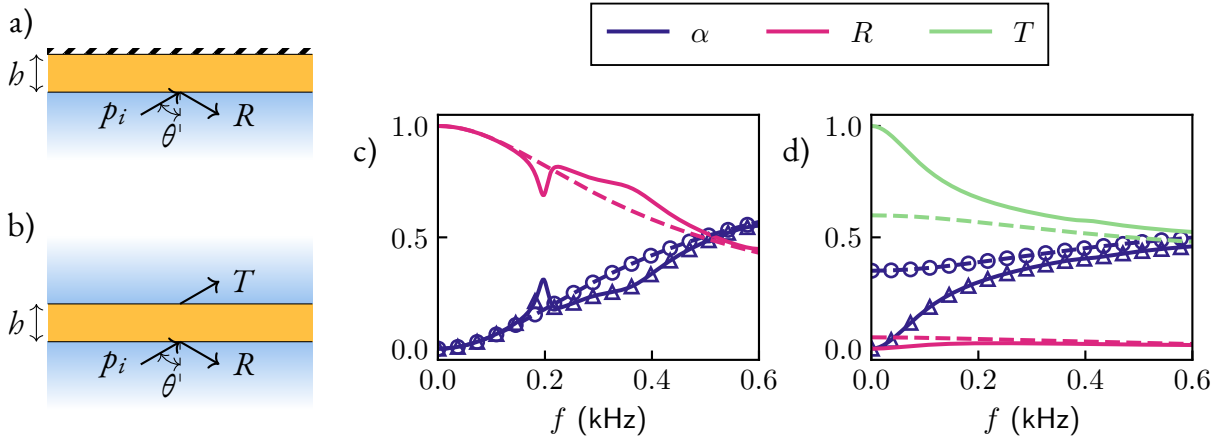


Fig. 1.3: Diagram for the a) reflection, b) transmission problem. Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the c) reflection, d) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid ($\circ$) and for the absorption coefficient (Δ)

or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve wavenumbers.

Computing the modes with semi-analytical tools

dispersion in porous⁸² + groby2008 +

Once the linear system is obtained, we are faced with either or two of these possibilities. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible.

Using Eq. (1.18), root finding such that $\det(\mathcal{M}(k_1, \omega)) = 0$; the determinant of the matrix from Eq. 1.17 can be expressed as

$$2 \frac{k_{2,eq} k_{2,0}}{\rho_{eq} \rho_0} \cos(k_{2,eq} h) + i \left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2} \right) \sin(k_{2,eq} h) = 0 \quad (1.20)$$

Numerics for mode computations

Regarding the computation of complex modes in homogeneous layer, several numerical methods have been investigated in the last decades. Guided waves represents a large topic and we focus here on the case of simple geometries; similar configurations are considered, an unbounded single (or stack of) layers with two infinite interfaces as presented in the section where we were interested in their acoustic response.

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM), which has been especially used for the computation of modes for guided waves in layers.⁹²

In the case of such simple geometries, the use of finite elements seems farfetched but could be relevant in situations where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include [lister des publis ici](#). Examples are also given in Chapter 2. A detailed description of the calculation of Lamb modes using a simple

Fig. 1.4: Discretization for each method

SAFE implementation using high-order Lobatto polynomials is proposed in Appendix. ???. There, its performance is also compared to that of SCM.

A bestiary of the wave solutions in dissipative and open waveguides

- **conservative systems**: simple waveguide: guided and trapped waves and purely evanescent
- **non-dissipative leaky**: still a waveguide but now an open system, waves are thus attenuated and radiate energy to the open domain. They are thus considered quasi-guided since they are propagating waves with some attenuation. Trapped modes can still occur. New family of modes appear as leaky backward and leaky forward waves that exhibit unique properties. Their counterparts are incoming waves.
- **dissipative leaky**: same but no trapped modes exist ? + séparation impossible entre incoming et leaky waves
- leaky modes, energy radiation: relevance of the imposition of a local coupling contrary to a full modelling of the surrounding media with some termination condition satisfying the Sommerfeld condition, PML or analogous.
- discuss surface and guided waves

1.2.4. Numerics for the computation of dispersion relations

- presentation of scm, safe + possible with fem
- cite chap. 2

1.3 | Methods for modelling metaporous structures

1.3.1. Scattering of waves by periodic structures

As has been studied in the introduction, a large literature exists showing the enhancement of absorption properties by periodicizing the porous layers with embedded scatterers. Periodicity can be imposed in acoustic systems using the so-called Bloch-Floquet theory. There, the layer of infinite extent is limited to a single unit-cell, able to alone describe the full behaviour of the infinitely repeating periodic cell.

- reciprocal wavenumber space, limit the study to the first Brillouin zone covering, $k_1 \in [0, \pi/d]$.
- opening of a Bragg bandgap: conditions of existence in classical metamaterial, can it happen with losses.
- With these complex geometries comes the need for more advanced tools for the discretization of these structures. Introduce finite elements.
- soft solids: aberkane phd

Media periodization

For more complex geometries, we must turn to alternative methods. As the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

Paragraphe periodic systems

To introduce periodicity in the configurations we have been investigating up to this section, we will first introduce a few fundamental concepts. Periodic structures and lattices are the central topic of crystallography which has been used in various disciplines such as condensed matter. In wave physics, periodic structures have been shown to introduce interesting effects towards applications for wave control and constitute the cornerstone of metamaterials as was shown in the literature presentation in Section [cite sec.](#)

For wave phenomena, periodicity was formalised by the Bloch-Floquet theorem. Taking an infinitely repeating lattice or a crystal, the description of the wave propagation in only one of the so-called unit cell, one spatial period of the total structure, is enough to describe the whole structure. A wave function $\psi(x)$ describing the propagation should thus depend on a spatially periodic field, with the same periodicity as the crystal modulated by the Bloch wavenumber k_q , such that,

The scattering of waves by a single unit cell is expressed using a wavefunction that is projected onto an infinite modal basis, the Bloch mode basis.

$$\psi_q(x) = e^{ik_q x}. \quad (1.21)$$

Their value at the left boundary is thus related to that at the right boundary modulated by the spatial periodicity and some translation vector. This formalism forms the fundamental framework for periodic structures in acoustics. These structures have been shown to support bands of frequency where waves are unable to propagate: band gaps. These occur at the Bragg frequency described as $d \sin(\theta) = n\lambda$ corresponding to destructive interference for an incident wave impinging on the structure. There are various phenomena and ways to obtain different types of band gaps^{stuff here}.

Overview of the methods

Multiple Scattering Theory is an analytical formalism developed for this purpose and is able to account for the scattering of each of the neighbouring obstacles. Tedious theoretical framework, shown to compute the acoustic response of metaporous⁷⁰ and metaporoelastic surfaces⁷⁸.

FEM flexible and powerful able to

1.3.2. Dispersion relations and band structure in periodic lattices

Most popular methods for the computation of band structure in 2D periodic systems made of classical lossless materials are the Plane Wave Expansion (PWE) which is semi-analytical. It relies on a formulation... Extended Plane Wave Expansion (EPWE) was generalized to obtain evanescent Bloch modes in Ref.^{schalcher2023}, allowing for complex dispersion diagram investigation. The complexity of the system prohibits the use of such semi-analytical methods. Multiple scattering could also be considered. However, it has been successfully used to describe the acoustic scattering of periodically arranged rigid inclusions in equivalent fluid unit cells⁷⁰ as well as solid inclusions in a poroelastic unit cell⁷⁸. However, the so-called Schlömilch series that appear in the calculation, from the effect of the scattering of the neighbouring cylindrical inclusions appear to be not so manipulable when considering complex longitudinal wavenumbers. Its expression is

$$\sigma_q^X = \sum_{l=1}^{\infty} H_q^{(1)}(k^X l d) \left((-1)^q e^{ik_1^X l d} + e^{-ik_1^X l d} \right), \quad (1.22)$$

and indeed, separating the k_1^X real and imaginary inevitably leads to at least one growing exponential term in the series, leading to it being unable to converge, up to our study.

The wfe as well as FEM has also been widely applied to these geometries. Their modularity and versatility lead them to be good candidates for the modelling of our structure.

We face challenges when trying to model this structure using the tools classically used in the literature, that is COMSOL Multiphysics. Although there are built-in tools in the software for "*Eigenfrequency*" studies, the library aims at solving complex frequencies of the system, given a set of parameters for the wavenumber. As it has been discussed earlier in the chapter, the goal of this work is to solve for complex wavenumbers and real frequencies. Even trying a manual implementation of the model in the "*Weak Form PDE*" interface, the tendency of this software to function as a black box introduces a new degree of complexity to this strategy. While FEM remains the soundest strategy for the modeling of the complex inhomogeneous geometry of the unit cell of metaporous surface, we propose to adapt in-house existing frameworks in order to develop numerical tools in this thesis.

A general spectral collocation method for computing the dispersion relations of guided acoustic waves in multilayer dissipative structures

Contents

2.1	Abstract	21
2.2	Introduction	21
2.3	Solving the dispersion relations for multilayer structures	22
2.3.1	Spectral Collocation Method	22
2.3.2	Solving the dispersion relation with Müller algorithm	25
2.4	Application to a two-layer poroelastic-elastic structure	26
2.4.1	Numerical scheme	26
2.4.2	Dispersion relation	29
2.4.3	Mode shapes and energy flow	30
2.4.4	Experimental validation	31
2.5	Conclusion	33
2.6	Appendices	34
2.6.A	Expression of the Biot elastic coefficients and effective parameters in poroelastic materials	34
2.6.B	Application to a single poroelastic layer surrounded by two fluid half-spaces	35
2.6.C	Evaluation of the physical fields from the SCM solutions	36
2.6.D	Energy flux in the multilayer structure - Poynting theorem in a poroelastic medium	37
2.6.E	Reminder of the SLaTCoW method for obtaining complex-valued wavenumbers.	37

THIS CHAPTER IS BASED ON THE MATERIAL PUBLISHED IN^{P1}, AUTHORED BY *MATHIEU MARÉCHAL, ALAN GESLAIN, JEAN-PHILIPPE GROBY, VICENT ROMERO-GARCÍA AND OLIVIER DAZEL*. MINOR MODIFICATIONS TO THE TEXT AND FIGURES HAVE BEEN MADE TO FIT THE FORMAT AND NOTATION OF THIS THESIS¹

¹While the remaining of the manuscript assumes a positive harmonic time-dependance, the reader should bear in mind that this entire chapter is written using a negative time-dependance, i.e. $e^{-i\omega t}$.

2.1 | Abstract

A spectral collocation method is proposed to compute the complex wavenumber-real frequency dispersion relations of guided acoustic waves in multilayer structures involving dissipative materials. The nature of these dissipative materials is initially considered to be arbitrary, i.e., poroelastic, viscoelastic, or viscoacoustic. For a given frequency, the complex wavenumbers as well as the physical fields, which are further used to evaluate the Poynting vectors and analyze the energy flux, are obtained by solving a generalized eigenvalue problem. The latter arises from a set of discretized equations of motion and appropriate boundary (coupling) conditions. These equations of motion and boundary (coupling) conditions are imposed by the nature of the material composing each layer of the structure. A focus is made on poroelastic layers. The dispersion relation of a two-layer elastic-poroelastic structure is analyzed, as well as the energy flows in the structure. The results as calculated with the present spectral collocation method are validated against those obtained with a classical complex root-finding (Müller) method and experiments.

2.2 | Introduction

Dealing with guided acoustic waves in multilayer structures of dissipative (viscoelastic, viscothermal, or poroelastic) material requires the evaluation of complex wavenumber-real frequency dispersion relations. Solving for these dispersion relations consists in finding complex roots of complex characteristic equations, which often arise from complex matrix determinants and can rarely be solved analytically. Various methods^{93,94}, among which the Müller algorithm⁹⁵, are commonly used to achieve this challenging task. These methods usually rely on preliminary semi-analytical derivations, require multiple initial guesses, and are iterative, so they can get stuck in local minima. Computing the complex wavenumber-real frequency dispersion relation in a reliable and robust way is thus of primary importance. Commercial software such as Disperse⁹⁶ or open-source software such as Dispersion Calculator^{97,98} are available for calculating dispersion curves. They use mode tracing or root-finding methods and focus mainly on multilayer elastic structures. Alternatively, several numerical methods can be used, such as the Finite Element Method⁹⁹, the Semi-Analytical Finite Element method (SAFE)^{100–102}, or spectral methods¹⁰³. Spectral methods are especially relevant in this framework. They are numerical approximation techniques that seek the solution of a differential equation using a finite series of infinitely differentiable basis functions with *spectral accuracy*⁹².

The propagation and dispersion of waves in multilayer systems have long been investigated^{104,105}. Spectral methods have been developed to cope with wave radiation problems, but have also emerged in the last decades to solve dispersion relations¹⁰⁶. Effectively, the numerical scheme boils down to discretizing the geometry along the layer thickness and solving an eigenvalue problem. Here, a collocation method is used to discretize the geometry. A finite-dimension polynomial basis is chosen and the collocation points are distributed according to the roots of this polynomial basis. Leaky modes appear in practice when multilayer structures are surrounded by one or two half-spaces. Accounting for this phenomenon brings us closer to practical considerations regarding these systems, as they would radiate part of the acoustic energy back to the surrounding medium, although it adds hardships to the modeling of the system; the waves modeled in these half-spaces should satisfy some radiation condition. These can be imposed by implementing perfectly matched layers¹⁰⁷, by using a bounded mapping, and applying a Spectral Collocation Method (SCM) to all layers of the extended system¹⁰⁸, or by using a suitable change of variable and solving a non-linear eigenvalue problem¹⁰⁹. This last method inspires the current work. We propose a SCM suitable for solving complex wavenumber-real frequency dispersion relations in multilayer systems involving dissipative

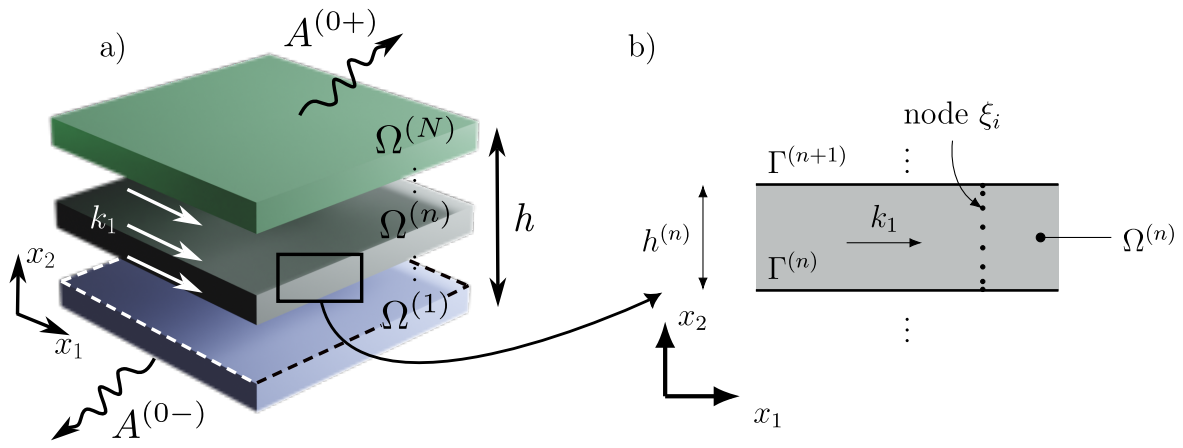


Fig. 2.1: a) Sketch of the geometry of the multilayer system and b) zoom on the n -th layer.

media. As a way of example, we will focus on multilayer poroelastic systems. Mechanical wave propagation in bulk poroelastic materials are well understood and modeled thanks to the seminal contributions of M.A. Biot^{88,89,110,111} and subsequent contributions, notably by D.L. Johnson *et al.*¹¹² and J.-F. Allard *et al.*^{19,113} concerning the modeling of viscothermal losses. Although poroelastic materials are bounded in practice and are usually encapsulated in more complex multilayer structures combining layers of materials of different nature in geophysics and acoustics, there are only a few studies on the dispersion relation of guided acoustic waves in poroelastic^{78,82,114}, porous^{groby2008a, 115} or anisotropic poroelastic layers¹¹⁶. The SCM has already been used to solve the dispersion relation of guided acoustic waves in elastic layered rings^{117,118} or anisotropic elastic multilayer systems¹¹⁹, but has never been used to cope with multilayer systems involving dissipative materials of arbitrary nature.

This work aims at filling this gap and is divided into two sections. First, the SCM is derived for generic multilayer configurations possibly coupled to two surrounding identical fluid half-spaces. A two-layer elastic-poroelastic structure is then considered by way of example. The dispersion relation computed with the SCM method is compared to that obtained with a classical root-finding (Müller) method and that measured. The energy flow is further analyzed for this two-layer configuration, with very little additional computational cost. In the last section, we experimentally analyze the two-layer structure by retrieving its complex dispersion relation from displacement measurements using the SLaTCoW (Spatial LAPlace Transform for COMplex Wavenumber recovery). The previously obtained numerical results can thus be correlated with the data of the dispersion relation.

2.3 Solving the dispersion relations for multilayer structures

2.3.1. Spectral Collocation Method

Figure 2.1a depicts the general geometry of the multilayer structure considered in this article. It is composed of N homogeneous layers of arbitrary nature. The n -th layer has a thickness of $h^{(n)}$ and occupies the domain $\Omega^{(n)}$. The lower and upper interfaces are denoted $\Gamma^{(n-1)}$ and $\Gamma^{(n)}$. The multilayer structure is assumed to be two-dimensional, of infinite extent along x_1 , and bounded along the x_2 -axis between 0 and the total structure thickness $h = \sum_{n=1}^N h^{(n)}$. Outgoing waves radiate above and below the structure in two fluid half-spaces $\Omega^{(0\pm)}$, occupied by the same fluid.

We assume an implicit time dependence $e^{-i\omega t}$. The physical fields associated to the guided waves

in the n -th layer propagating along the positive x_1 -axis take the form

$$\Theta^{(n)}(x) = s^{(n)}(k_1, x_2) e^{ik_1 x_1}, \quad (2.1)$$

where $s^{(n)}(k_1, x_2)$ is the spatial Laplace transform of $\Theta^{(n)}(x)$ and k_1 is the complex valued x_1 -component of the wavenumber. Note that the spatial Laplace transform is used instead of the spatial Fourier transform because k_1 is complex-valued. because k_1 , the x_1 component of the wavenumber is complex-valued. The physical fields that compose $\Theta^{(n)}(x)$ depend on the nature of the material occupying the n -th layer. Introducing $\Theta^{(n)}(x)$ in the corresponding wave equations (of second order in space in this work) reduces to a second-order differential operator $\mathcal{L}^{(n)}$ in x_2 , that depends on the material properties and can further be developed as a second-order polynomial in k_1

$$\mathcal{L}^{(n)}(x_2) s^{(n)}(k_1, x_2) = \left(k_1^2 \mathcal{L}_2^{(n)}(x_2) + k_1 \mathcal{L}_1^{(n)}(x_2) + \mathcal{L}_0^{(n)}(x_2) \right) s^{(n)}(k_1, x_2) = 0, \quad (2.2)$$

where $\mathcal{L}_j^{(n)}$ are the coefficients of the polynomial expansion $j = 0, 1, 2$. Please note that this operator is nothing but the spatial Laplace transform of the equations of motion.

The interface (boundary) conditions at $\Gamma^{(n)}$ between the $n+1$ -th and the n -th layers again depend on the nature of these two layers, but can be formally written in the form

$$C^{(n)-} s^{(n)}(k_1) \Big|_{\Gamma^{(n)}} - C^{(n)+} s^{(n+1)}(k_1) \Big|_{\Gamma^{(n)}} = 0, \quad (2.3)$$

where $C^{(n)\pm}$ are interfaces operators, which involve differential operators of maximum second-order in x_2 . Following this procedure and introducing $S = \left(s^{(0-)} \dots s^{(n)} \dots s^{(0+)} \right)^T$, with T denoting transposition, the problem can be cast in matrix form, whose coefficients are differential operators in x_2 ,

$$K S(k_1, x_2) = \begin{pmatrix} C^{(0-)} & -C^{(0+)} & 0 & & & \\ 0 & \mathcal{L}^{(1)}(x_2) & 0 & & & \\ 0 & C^{(1-)} & -C^{(1+)} & 0 & & \\ & 0 & \mathcal{L}^{(2)}(x_2) & 0 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 0 & \mathcal{L}^{(N)}(x_2) & 0 \\ & & & 0 & C^{(N-)} & -C^{(N+)} \end{pmatrix} S(k_1, x_2) = 0. \quad (2.4)$$

Note that the equations of motion in both upper and lower half-spaces are not required in Eq.(2.4). This system only depends on the values of the two spatial Laplace transforms $s^{(0\pm)}$ at the upper and lower interfaces. Instead, we specify the nature of the two half-spaces, i.e. a fluid medium, and make use of the Sommerfeld condition, to explicitly give the form of the spatial Laplace transforms $\tilde{p}^{(0\pm)}(k_1, x_2)$ of the pressure fields $p^{(0\pm)}(x)$

$$\begin{aligned} \tilde{p}^{(0-)}(k_1, x_2) &= A^{(0-)} e^{-ik_2^{(0)} x_2}, \in \Omega^{(0-)}, \\ \tilde{p}^{(0+)}(k_1, x_2) &= A^{(0+)} e^{ik_2^{(0)}(x_2-b)}, \in \Omega^{(0+)}, \end{aligned} \quad (2.5)$$

with $k_2^{(0)} = \sqrt{(k^{(0)})^2 - k_1^2}$, such that $\text{Re}(k_2^{(0)}) \geq 0$, where $k^{(0)} = \omega/c_f$ is the wavenumber in the fluid and c_f the wave velocity. Thus, $s^{(0\pm)}$ appearing in Eq.(2.4) reduces to $s^{(0\pm)} = A^{(0\pm)}$. $C^{(0-)}$ and $C^{(N+)}$ are therefore no longer operators and reduce to $C^{(0-)} = C^{(0-)}$ and $C^{(N+)} = C^{(N+)}$. In our fluid case, the interface conditions at $\Gamma^{(0)}$ and $\Gamma^{(N)}$ involve the spatial Laplace transform of the

pressure and its first spatial derivative with respect to x_2 whatever the nature of the 1-st and N -th layers. Both matrices $C^{(j)}$, $j = (0^-), (N^+)$, can thus be formally written as

$$C^{(j)} = k_1 C_1^{(j)} + C_0^{(j)} + k_2^{(0)} C_{1'}^{(j)}, \quad (2.6)$$

with the subscript denoting the submatrices rearranged according to the power order k . Index $1'$ corresponds to the term in $k_2^{(0)}$. Introducing Eqs.(2.2) and (2.6) in Eq. (2.4) leads to

$$\left(k_1^2 K_2 + k_1 K_1 + K_0 + i k_2^{(0)} K_{1'} \right) S = 0. \quad (2.7)$$

This eigenvalue problem is non-linear because of the presence of the term $k_2^{(0)}$. Making use of the following changes of variable $k_1 = k^{(0)} (\gamma + \gamma^{-1}) / 2$ and $k_2^{(0)} = k^{(0)} (\gamma - \gamma^{-1}) / 2i$ ^{120,121} and companion linearization¹²², a generalized eigenvalue problem is formed and solved for γ , from which the dispersion relation can be calculated. More details on this procedure can be found in¹⁰⁹. Note that the first or the last column of Eq.(2.4) vanishes when radiation is absent on either side of the multilayer system. The situation is different when both fluid half-spaces disappear. The eigenvalue problem Eq.(2.7) is no longer non-linear and the problem can directly be solved after companion linearization. Note also that fluid half-spaces are only considered for the sake of simplicity and that this procedure can be adapted to any kind of material occupying the half-spaces, as described recently¹²³. Note finally that this procedure avoids discretization of these two half-spaces.

To numerically solve the problem, the spatial Laplace transform of the physical fields in each n -th layer are expanded as

$$s^{(n)}(x_2) \approx \underline{s}^{(n)}(x_2) = \sum_{m=0}^M \alpha_m^{(n)} \underline{\psi}_{-m}(\underline{\xi}^{(n)}), \quad (2.8)$$

where α_m are coefficients of the polynomial expansion, $\underline{\psi}_{-m}$ are the Chebyshev polynomial of order m , and $\underline{\xi}^{(n)} = 2(x_2 - \sum_{j=1}^{n-1} h^{(j)}) / h^{(n)} - 1$, $\in [-1, 1]$. Underlined variables represent discrete vectors and double-underlined variables discrete matrices. The goal of this approximation is to discretize the differential operators in x_2 appearing in the problem. To do so, the SCM is employed. $\underline{\xi}^{(n)}$ is discretized on the roots of the Chebyshev polynomials, $\underline{\xi}_{-j}^{(n)} = \cos\left(\frac{j\pi}{M^{(n)}}\right)$ with $j = 0 \dots M^{(n)}$. Please note that $\underline{\xi}_{-j}^{(n)}$ discretely runs from 1 to -1 with increasing j , while $\underline{\xi}^{(n)}$ continuously runs over the same interval but in the opposite direction, i.e. from -1 to 1. These collocation points/nodes form a non-uniform grid along the thickness of each layer. This discretization is usually preferred because the collocation points are clustered at the interfaces⁹². The locations of these nodes are presented in Fig. 2.1b for the n -th layer with an arbitrary value $M^{(n)} = 8$. The differential operators in the n -th layer are thus represented by differentiation matrices (DMs), such that

$$\begin{aligned} s^{(n)}(x_2) &\rightarrow \underline{I} \underline{s}^{(n)}(\underline{\xi}_{-j}), & \partial_2 s^{(n)}(x_2) &\rightarrow \left(\frac{2}{h^{(n)}} \right) \underline{\underline{D}}_2 \underline{s}^{(n)}(\underline{\xi}_{-j}) = \underline{\underline{D}}_2^{(n)} \underline{s}^{(n)}(\underline{\xi}_{-j}), \\ \partial_{22} s^{(n)}(x_2) &\rightarrow \left(\frac{2}{h^{(n)}} \right)^2 \underline{\underline{D}}_{22} \underline{s}^{(n)}(\underline{\xi}_{-j}) = \underline{\underline{D}}_{22}^{(n)} \underline{s}^{(n)}(\underline{\xi}_{-j}). \end{aligned} \quad (2.9)$$

where $\underline{I}$ is the identity matrix and $\underline{\underline{D}}_2$ and $\underline{\underline{D}}_{22}$ are respectively the first-order and second-order normalized DMs. Contrary to an interpolation with Chebyshev polynomials, DMs interpolate the functions and their spatial derivatives directly and thus do not involve the polynomial expansion coefficients per se.¹²⁴ Note that the normalized DMs are denoted as $\underline{\underline{D}}_2^{(n)}$ and $\underline{\underline{D}}_{22}^{(n)}$ in the following. These matrices are square and have a size of $(M \times M)$. This size also directly depends on the order

of the considered Chebyshev polynomials. The formulation prevents the occurrence of roundoff errors, which is of particular importance when SCM is applied to solve eigenvalue problems¹²⁵.

The linear operator $\mathcal{L}^{(n)}$ is discretized as a matrix $\underline{\underline{L}}^{(n)}$ with size $(P^{(n)}(M^{(n)} - 1), P^{(n)}(M^{(n)} + 1))$, where $P^{(n)}$ is the number of physical fields required to describe the wave propagation in the layer and again depends on the nature of the material this layer is composed of. The first and the last rows of the DMs are removed because they are used to apply the interface conditions at $\Gamma^{(n+1)}$ and $\Gamma^{(n)}$ respectively. The interface operators $C^{(n)\pm}$ are effectively expressed in terms of this first row of the DMs of the n -th layer and this last row of the DMs of the $(n + 1)$ -th layer. These rows respectively correspond to the first, i.e. $\xi_{M^{(n)}} = -1$ and $\xi^{(n)} = 1$, and last collocation nodes of the n -th and $(n + 1)$ -th layers, i.e. $\xi_0 = 1$ and $\xi^{(n+1)} = -1$. These vectors are subsequently denoted $\underline{I}^{(n+1)-}$, $\underline{I}^{(n)+}$, $\underline{D}_2^{(n+1)-}$, $\underline{D}_2^{(n)+}$, $\underline{D}_{22}^{(n+1)-}$, and $\underline{D}_{22}^{(n)+}$. The interface matrix $\underline{\underline{C}}^{(n)-}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n)}(M^{(n)} + 1))$ and the interface matrix $\underline{\underline{C}}^{(n)+}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n+1)}(M^{(n+1)} + 1))$.

Finally, the full $(\sum_{n=0}^N M^{(n)} P^{(n)} + 2, \sum_{n=0}^N M^{(n)} P^{(n)} + 2)$ -matrix system for the general multilayer system is

$$\underline{\underline{K}} \underline{\underline{S}} = \begin{pmatrix} \underline{\underline{C}}^{(0-)} & -\underline{\underline{C}}^{(0+)} & 0 & & & \\ 0 & \underline{\underline{L}}^{(1)} & 0 & & & \\ 0 & \underline{\underline{C}}^{(1-)} & -\underline{\underline{C}}^{(1+)} & 0 & & \\ & 0 & \underline{\underline{L}}^{(2)} & 0 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 0 & \underline{\underline{L}}^{(N)} & 0 \\ & & & 0 & \underline{\underline{C}}^{(N-)} & -\underline{\underline{C}}^{(N+)} \end{pmatrix} \begin{pmatrix} \underline{A}^{(0-)} \\ \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ \vdots \\ \underline{s}^{(N)} \\ \underline{A}^{(0+)} \end{pmatrix} = \underline{0}. \quad (2.10)$$

This system is cast in the form of a generalized eigenvalue problem that is solved by traditional eigenvalue solvers at each frequency. The dispersion relation for the acoustic guided wave in the dissipative multilayer system is calculated, repeating the procedure for each frequency ω .

2.3.2. Solving the dispersion relation with Müller algorithm

The dispersion relation of multi-layer dissipative structure are more commonly calculated with secant-based algorithm as the Müller algorithm. The latter relies on a semi-analytic description of the fields. These fields are best written in each layer, independently of the nature of the material it is composed of, using the potentials, i.e. decomposing the field $\theta = \nabla \varphi + \nabla \times \psi$ into an irrotational component φ and a transverse component $\psi = \psi \mathbf{e}_3$. The spatial Laplace transform of the potentials in the n -th layer are entirely described by

$$\tilde{\theta}_n(k_1, x_2) = A_X^{(n)+} e^{ik_{2,X}^{(n)}(x_2 - b^{(n)})} + A_X^{(n)-} e^{-ik_{2,X}^{(n)}(x_2 - b^{(n)})}, \quad (2.11)$$

with $A_X^{(n)\pm}$ the up and down-going wave amplitudes, $k_{2,X}^{(n)} = \sqrt{(k_X^{(n)})^2 - k_1^2}$, such that $\text{Re}(k_{2,X}^{(n)}) \geq 0$, and X refers to the type of wave, i.e. shear, compressional or acoustic. If necessary, these Laplace transform representations are complemented by those of the pressure field in the fluid half-spaces provided in Eq.(2.5).

The boundary conditions are applied at each interface and the set of equations is cast into the following matrix form $\mathbf{M}(k_1)\mathbf{q} = 0$, with $\mathbf{q}$ a vector containing the amplitudes of each field/potential. The modes of the system correspond to the wavenumbers k_1 for which the matrix $\mathbf{M}(k_1)$ is singular, i.e.

$$\det(\mathbf{M}(k_1)) = 0. \quad (2.12)$$

These wavenumbers are the complex roots of an implicit complex-valued equation. They are evaluated by running our own implementation of the Müller algorithm⁹⁵ from 3 initial guesses. This equation is solved iteratively for each frequency ω , giving the dispersion relation.

In addition, $\mathbf{q}$ can be obtained by computing the nullspace of $\mathbf{M}$ ¹²⁶ (This null space is computed using singular value decomposition as embedded in the `linalg.null_space` function from the Python package SciPy). The amplitudes of the potentials in the layers are thus evaluated, enabling all the physical fields that depend on these potentials to be calculated. However, the $\mathbf{M}$ matrix becomes increasingly tedious to write as the number of layers increases, and even more so when it comes to calculating the roots of its determinant and nullspaces. Nevertheless, this approach remains a fairly effective way of providing a reference solution for validating numerical results when studying a two-layer system.

2.4 Application to a two-layer poroelastic-elastic structure

In this section, the calculation of the dispersion relation for guided waves in a two-layer structure, i.e. $N = 2$, is used as an example of the application of the present SCM. This two-layer structure consists of a $h^{(1)}$ -thick poroelastic layer, coated on one side by a $h^{(2)}$ -thick aluminum plate, that is radiating in a fluid half-space, and rigidly backed on the other side as depicted in Fig. 2.2a. The SCM results are analyzed and compared to those obtained with the Müller root-finding method.

2.4.1. Numerical scheme

The pressure field $p^{(1)}$ and the two components of the frame displacement, $u_{1,s}^{(1)}$ and $u_{2,s}^{(1)}$, i.e. $P^{(1)} = 3$, as well as the two-components of the elastic displacement, $u_1^{(2)}$ and $u_2^{(2)}$, i.e. $P^{(2)} = 2$, are used to model the propagation in the poroelastic and in the elastic layers respectively. The $\{\mathbf{u}_s, p\}$ formulation is used⁹⁰ for the poroelastic medium. The more usual displacement formulations ($\{\mathbf{u}_s, \mathbf{u}_f\}$ ⁸⁸ or $\{\mathbf{u}_s, \mathbf{w}\}$ ¹¹⁰) cannot be used in the present case, because the number of physical fields they rely on, i.e. 4, is different from that of the potentials in the layer, i.e. 3.

In total, 5 physical fields inside the two-layer structure and an additional amplitude for the acoustic wave radiated towards $x_2 \rightarrow \infty$ are needed to solve for the dispersion relation. Each domain is discretized on $M^{(j)}$, $j = 1, 2$, collocation points.

In more details, the $\{\mathbf{u}_s, p\}$ formulation of the Biot theory⁹⁰ is employed to model the propagation in the homogeneous isotropic poroelastic layer. The coupled equations of motion read as

$$\begin{cases} \nabla \cdot \hat{\boldsymbol{\sigma}}_s^{(1)} + \hat{\rho} \omega^2 \mathbf{u}_s^{(1)} + \beta \nabla p^{(1)} = 0, \\ \frac{\nabla^2 p^{(1)}}{\rho_{22} \omega^2} - \frac{\beta}{\phi^2} \nabla \cdot \mathbf{u}_s^{(1)} + \frac{1}{R} p^{(1)} = 0, \end{cases} \quad (2.13)$$

where $\hat{\boldsymbol{\sigma}}_s^{(1)} = \hat{A} \nabla \cdot \mathbf{u}_s^{(1)} \mathbf{I} + 2N_s \boldsymbol{\epsilon}_s$ is the *in vacuo* solid stress tensor, with $\boldsymbol{\epsilon}_s$, the solid-phase strain tensor. The parameters are generally expressed in terms of the elastic coefficients P , Q and R and the effective densities ρ_{11} , ρ_{12} and ρ_{22} initially introduced by Biot^{11,88} as

$$\hat{A} = P - 2N_s - \frac{Q^2}{R}, \quad \hat{\rho} = \rho_{11} - \frac{\rho_{12}^2}{\rho_{22}}, \quad \beta = \phi \left(\frac{\rho_{12}}{\rho_{22}} - \frac{Q}{R} \right). \quad (2.14)$$

These parameters depend on the frame properties and on the effective density $\rho_{eq}(\omega)$ and bulk modulus $K_{eq}(\omega)$ of the fluid phase as reminded in 2.6.A. In short, a poroelastic material is entirely described by the porosity ϕ , the shear modulus of the frame N_s , the Poisson ratio ν , the density of

the frame ρ_1 , the viscous and thermal characteristic lengths Λ and Λ' , tortuosity α_∞ , and the flow resistivity R_f when saturated by a light fluid^{88,112,113}. All these parameters are real valued except for the shear modulus, which incorporates a viscoelastic damping factor.

Introducing $\underline{s}^{(1)} = (\tilde{u}_1^s, \tilde{u}_2^s, \tilde{p})^T$, the spatial Laplace transform of Eq. (2.13) is expanded in the form of Eq. (2.2) with

$$\begin{aligned} \mathcal{L}_0^{(1)} &= \begin{pmatrix} N_s \partial_{22} + \hat{\rho} \omega^2 & 0 & 0 \\ 0 & \hat{P} \partial_{22} + \hat{\rho} \omega^2 & \beta \partial_2 \\ 0 & -\beta / \phi^2 \partial_2 & 1/R + \partial_{22} / (\rho_{22} \omega^2) \end{pmatrix}, \\ \mathcal{L}_1^{(1)} &= \begin{pmatrix} 0 & i \hat{P} \partial_2 & i \beta \\ i(\hat{A} + N_s) \partial_2 & 0 & 0 \\ -i \beta / \phi^2 & 0 & 0 \end{pmatrix}, \text{ and } \mathcal{L}_2^{(1)} = \begin{pmatrix} \hat{P} & 0 & 0 \\ 0 & N_s & 0 \\ 0 & 0 & 1 / (\rho_{22} \omega^2) \end{pmatrix}. \end{aligned} \quad (2.15)$$

Finally, the discretized equations of motion read as

$$\underline{\underline{L}}^{(1)} \underline{\underline{s}}^{(1)} = \begin{pmatrix} (\omega^2 \hat{\rho} - k_1^2 \hat{P}) \underline{\underline{I}} + N_s \underline{\underline{D}}_{22} & i k_1 (\hat{P} - N_s) \underline{\underline{D}}_{22} & -i k_1 \beta \underline{\underline{I}} \\ i k_1 (\hat{P} - N_s) \underline{\underline{D}}_{22} & \hat{P} \underline{\underline{D}}_{22} + (\omega^2 \hat{\rho} - k_1^2 N_s) \underline{\underline{I}} & \beta \underline{\underline{D}}_{22} \\ -i \beta \omega^2 \underline{\underline{I}} & -\beta \omega^2 \underline{\underline{D}}_{22} & \frac{k_1^2}{\rho_{22}} \underline{\underline{I}} + \frac{\omega^2}{R} \underline{\underline{I}} \end{pmatrix} \underline{\underline{s}}^{(1)} = 0. \quad (2.16)$$

The equation of motion in the elastic medium is derived from the fundamental elasticity equations¹²⁷ and reads as,

$$\rho \omega^2 \underline{\underline{u}}^{(2)} + (\lambda + \mu) \nabla (\nabla \cdot \underline{\underline{u}}^{(2)}) + \mu \nabla^2 \underline{\underline{u}}^{(2)} = 0, \quad (2.17)$$

where ρ is the density and λ and μ are the Lamé coefficients. The corresponding discretized equation of motion is

$$\underline{\underline{L}}^{(2)} \underline{\underline{s}}^{(2)} = \left(\frac{\rho^{(2)}}{\mu} \omega^2 \underline{\underline{I}} + \begin{pmatrix} -k_1^2 (\lambda / \mu + 2) \underline{\underline{I}} + \underline{\underline{D}}_{22} & i k_1 (\lambda / \mu + 1) \underline{\underline{D}}_{22} \\ i k_1 (\lambda / \mu + 1) \underline{\underline{D}}_{22} & (\lambda / \mu + 2) \underline{\underline{D}}_{22} - k_1^2 \underline{\underline{I}} \end{pmatrix} \right) \underline{\underline{s}}^{(2)} = 0. \quad (2.18)$$

In addition, the same procedure can be followed for the equation of motion of a wave propagating in a layer of fluid material. We give its expression in order to provide an exhaustive list of the types of layers that can be modeled using the present method. The Helmholtz equation, $(\nabla^2 + k_0^2) p = 0$, with k_0 the bulk wavenumber and p the pressure field in the fluid material, is discretized into the form,

$$\left(\underline{\underline{D}}_{22} - k_1^2 + k_0^2 \right) \underline{\underline{p}} = 0. \quad (2.19)$$

The interface and boundary conditions depend on the vectors $\underline{\underline{I}}^\pm$, $\underline{\underline{D}}_2^\pm$ and $\underline{\underline{D}}_{22}^\pm$ which corresponds to the last (subscript +) and first row (subscript -) of the DMs. From the bottom to the top of the two-layer system, they are

— Rigid boundary condition at $\Gamma^{(0)}$ which leads to

$$\underline{\underline{u}}^{(1)}(x_2 = 0) = 0; \quad \underline{\underline{u}}_f^{(1)}(x_2 = 0) \cdot \underline{\underline{n}} - \underline{\underline{u}}_s^{(1)}(x_2 = 0) \cdot \underline{\underline{n}} = 0, \quad (2.20)$$

the discretized form of which is,

$$\underline{\underline{C}}^{(0)} \underline{\underline{s}}^{(1)} = \begin{pmatrix} 0 & \underline{\underline{I}}^+ \\ 0 & 0 \\ \frac{\phi}{\rho_{22}} \underline{\underline{D}}_2^+ & 0 \end{pmatrix} - \left(1 + \frac{\rho_{12}}{\rho_{22}} \right) \omega^2 \underline{\underline{I}}^+ \begin{pmatrix} \underline{\underline{p}}^{(1)} \\ \underline{\underline{u}}_1^{(1)} \\ \underline{\underline{u}}_2^{(1)} \end{pmatrix} = \underline{\underline{0}}. \quad (2.21)$$

- Continuity between the poroelastic and the elastic media¹²⁸ at $\Gamma^{(1)}$ which leads to

$$\begin{aligned} \sigma^{t(1)}(x_2 = b^{(1)}) - \sigma^{(2)}(x_2 = b^{(1)}) &= 0; \quad \mathbf{u}^{(2)}(x_2 = b^{(1)}) = \mathbf{u}_s^{(1)}(x_2 = b^{(1)}); \\ \mathbf{u}_f^{(1)}(x_2 = b^{(1)}) \cdot \mathbf{n} - \mathbf{u}_s^{(1)}(x_2 = b^{(1)}) \cdot \mathbf{n} &= 0, \end{aligned} \quad (2.22)$$

where $\sigma^{t(1)} = \hat{\sigma}_s - \phi(1 + Q/R)$ is the total stress tensor in the poroelastic layer. The corresponding discretized interface matrices, arising from

$$\underline{\underline{C}}^{(1)+} \underline{\underline{s}}^{(2)} - \underline{\underline{C}}^{(1)-} \underline{\underline{s}}^{(1)} = \underline{\underline{C}}^{(1)} \begin{pmatrix} \underline{p}^{(1)} & \underline{u}_1^{(1)} & \underline{u}_2^{(1)} & \underline{u}_1^{(2)} & \underline{u}_2^{(2)} \end{pmatrix}^T = 0, \quad (2.23)$$

are cast as

$$\underline{\underline{C}}^{(1)} = \left(\begin{array}{ccc|cc} -N_s \underline{D}_2^+ & -ik_1 N_s \underline{I} & 0 & \mu \underline{D}_2^- & ik_1 \mu \underline{I}^- \\ -ik_1 \hat{A} \underline{I}^+ & -\hat{P} \underline{D}_2^+ & \phi \left(1 + \frac{Q}{R} \right) \underline{I}^+ & ik_1 \lambda \underline{I}^- & (\lambda + 2\mu) \underline{D}_2^- \\ 0 & -\underline{I}^+ & 0 & \underline{I}^- & 0 \\ 0 & 0 & -\underline{I}^+ & 0 & \underline{I}^- \\ -\frac{\phi}{\rho_{22}} \underline{D}_2^+ & \left(1 + \frac{\rho_{12}}{\rho_{22}} \right) \omega^2 \underline{I}^+ & 0 & 0 & 0 \end{array} \right), \quad (2.24)$$

with the left-hand side corresponding to $\underline{\underline{C}}^{(1)-}$ and the right-hand side to $\underline{\underline{C}}^{(1)+}$.

- Continuity between the elastic and the fluid media at $\Gamma^{(2)}$ which leads to

$$\begin{aligned} \sigma_{12}^{(2)}(x_2 = b) &= 0; \quad \sigma_{22}^{(2)}(x_2 = b) = -p^{(0+)}; \\ \mathbf{u}_2^{(2)}(x_2 = b) &= \mathbf{u}_2^{(0+)}(x_2 = b). \end{aligned} \quad (2.25)$$

The corresponding discretized interface matrix $\underline{\underline{C}}^{(2)} \begin{pmatrix} \underline{u}_1^{(2)} & \underline{u}_2^{(2)} & A^{(0+)} \end{pmatrix}^T = 0$ is

$$\underline{\underline{C}}^{(2)} = \left(\begin{array}{cc|c} \underline{D}_2^+ & ik_1 \underline{I}^+ & 0 \\ ik_1 \lambda \underline{I}^+ & (\lambda + 2\mu) \underline{D}_2^+ & 1 \\ 0 & \omega^2 \underline{I}^+ & \frac{ik_2^{(0)}}{\rho_f} \end{array} \right). \quad (2.26)$$

Combining the discretized interface conditions, Eqs.(2.21), (2.24) and, (2.26), with the discretized equations of motion, Eqs.(2.16) and (2.18), leads to the following system

$$\begin{pmatrix} \underline{\underline{C}}^{(0)} & 0 & 0 \\ \underline{\underline{L}}^{(1)} & 0 & 0 \\ \underline{\underline{C}}^{(1)-} & -\underline{\underline{C}}^{(1)+} & 0 \\ 0 & \underline{\underline{L}}^{(2)} & 0 \\ 0 & \underline{\underline{C}}^{(2)-} & -\underline{\underline{C}}^{(2)+} \end{pmatrix} \begin{pmatrix} \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ A^{(0+)} \end{pmatrix} = \underline{0}. \quad (2.27)$$

This system is further expanded in the form of Eq.(2.7) and solved for a set of eigenvalues γ and associated eigenvectors $\mathbf{S}$ at each frequency ω . Wavenumbers k_1 are then obtained from γ . The dispersion curve is thus calculated over a given frequency range. However, the condition $\text{Re}(k_2^0) \geq 0$ should be imposed to sort out the solution that do not meet it.

Although usual and simple, this two-layer structure is challenging from a numerical point of view because it involves materials with very large impedance contrasts and layers whose thicknesses differ by several orders of magnitude. These problems are partially solved by normalizing each line and rearranging the terms of the matrix Eq. (2.7), leading to a better numerical conditioning of the system.

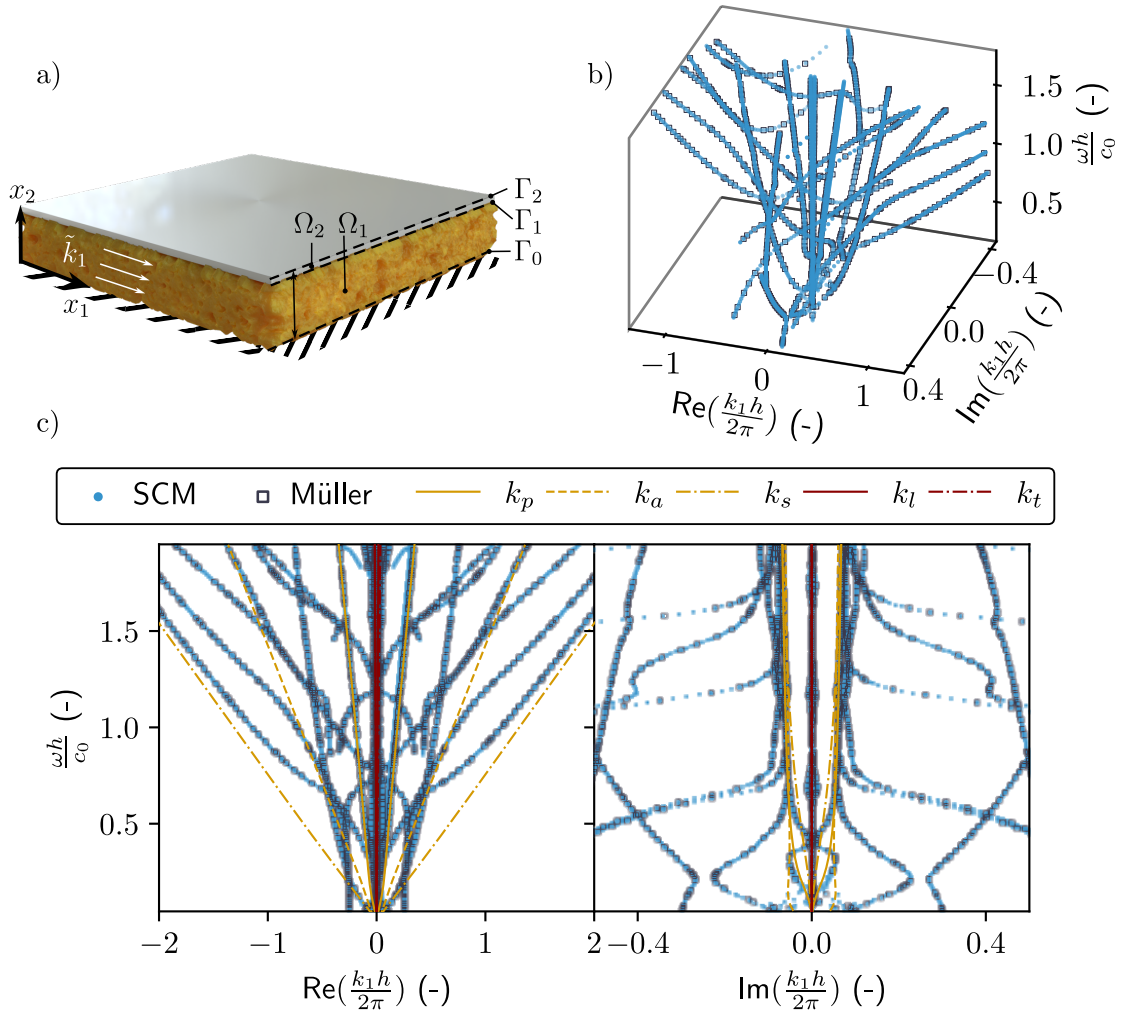


Fig. 2.2: a) Sketch of the two-layer structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. Blue dots represents the results as calculated with the SCM and black open markers represents those as calculated with the Müller method. Bulk wavenumbers in the poroelastic and elastic materials are highlighted in yellow and dark red respectively.

2.4.2. Dispersion relation

The material properties of the $h^{(1)} = 52$ mm-thick poroelastic (melanime) layer are listed in Table 2.1, while the properties of the $h^{(2)} = 1$ mm-thick aluminium plate are the density $\rho = 2700 \text{ kg.m}^{-3}$ and the Lamé coefficients $\lambda = 60.75 \text{ GPa}$ and $\mu = 26.03 \text{ GPa}$. The saturating fluid and that occupying the half-space is air with density $\rho_f = 1.213 \text{ kg.m}^{-3}$, the heat capacity ratio $\kappa = 1.4$, kinematic compressibility $\mu_f = 1.839 \times 10^{-5} \text{ Pa.s}$, the Prandtl number $\text{Pr} = 0.71$, and adiabatic bulk modulus κP_0 , where the atmospheric pressure is $P_0 = 1.013 \times 10^5 \text{ Pa}$. The total two-layer thickness is thus $h = h^{(1)} + h^{(2)} = 53 \text{ mm}$. $M^{(1)} = 5$ and $M^{(2)} = 11$ collocation points are employed respectively in the poroelastic and elastic layers.

The 3D view of the complex dispersion diagram is depicted Fig. 2.2b. To ease readability, real and imaginary parts are depicted separately in Fig. 2.2c. Please note that this last representation can be misleading because only the modes that lie in the complex wavenumber range depicted in Fig. 2.2b are represented in Fig. 2.2c. Cut-on frequencies are thus fictitious. The results calculated using the SCM method, plotted in blue markers, are compared with those obtained using the Müller method, plotted with black open markers. Both methods provide identical results, up to around four digits, therefore validating the present approach. Note that the SCM solutions are used as initial guesses of the Müller algorithm to shorten computational time. In this way, fewer iterations are needed for

Table 2.1: Parameters of the poroelastic material (melamine foam)

Parameter	ϕ	ρ_1	R_f	Λ	Λ'	α_∞	ν	N_s
Unit	-	kg.m ⁻³	kPa.s.m ⁻²	μm	μm	-	-	kPa
Melamine	0.98	6.5	5.6	214	214	1	0.24	11.96 (1 + i0.07)

the results of the root-finding method to converge. The main advantages of SCM are that no initial guesses are needed, i.e. as long as the discretization is sufficient to model the full wave behavior in the structure, the dispersion relation is guaranteed to be calculated. In addition, the matrices used for the numerical model are very small compared to those required by other methods such as FEM, and are therefore much less demanding in terms of calculation.

Wavenumbers are perfectly symmetric with both $\text{Re}(k_1 h/2\pi) = 0$, and $\text{Im}(k_1 h/2\pi) = 0$ axis, which is a feature of reciprocal systems. Wavenumbers having a very large slope and a low imaginary part correspond to modes mostly propagating in the aluminium plate. They are very close to the bulk longitudinal k_l and transverse k_t wavenumbers in the aluminum. This is due to the very large contrast between the elastic properties of the aluminum and those of the poroelastic frame. The other branches asymptotically tend toward the bulk acoustic, k_a , compression, k_p , or shear, k_s , wavenumbers of the poroelastic medium at high frequency. At low frequencies, these modes are highly dispersive.

If we compare these results with the dispersion relation depicted in 2.6.Bc) for the guided waves in a single poroelastic plate twice as thick and in the absence of the elastic plate, we find some essential differences. The branches associated with the elastic plate and the branch clearly associated with the coupling between the elastic and poroelastic layers (referenced as (1) in Fig.2.6.Bc)) do not exist. The other branches are similar to those of this single poroelastic layer, although slightly modified by the presence of the elastic plate. This proves a weak coupling between the elastic and poroelastic layers, particularly at high frequencies. At lower frequencies, the coupling between the two layers is stronger. Effectively, the wavelength is much larger than the thickness of at least one of the two layers in these frequency ranges, which favors coupling. The A_0 branch observed in the case of a single poroelastic layer disappear in the two-layer system rigidly backed, because this rigid-backing does not allow the existence of such a mode.

2.4.3. Mode shapes and energy flow

Beyond the SCM numerical efficiency, a significant advantage of the method is that the eigenvectors associated to each eigenvalue directly provides the associated mode shape. Since these mode shapes, i.e. eigenvectors, are calculated from an eigenvalue problem, their amplitude are defined up to a constant. They are normalized to the value of the pressure field at the bottom of the poroelastic layer, i.e. at the location of the rigid backing, because this condition imposes $\tilde{u}^{(1)}(0) = 0$ (and in particular $\tilde{u}_2^{(1)}(0) = 0$), which is equivalent to a maximum pressure field. From these normalized eigenvectors, any physical fields can be calculated as detailed in 2.6.C.

Let us consider two solutions $k_1^\pm = \pm |\text{Re}(k_1)|$ symmetric in the dispersion diagram with respect to the axis $\text{Re}(k_1) = 0$ as shown in Figs. 2.3a-b. The displacement fields and total energy fluxes in each direction are depicted in Fig. 2.3c-f. The results calculated using the SCM method, plotted in continuous lines, match those obtained using the Müller method, plotted with marker lines. These two modes are identical but propagate in opposite directions, i.e., the mode k_1^+ propagates towards positive x_1 while the mode k_1^- propagates towards negative x_1 . Displacement fields and energy fluxes are identical along the x_2 axis but are of equal modulus but opposite sign along the x_1 axis (see Fig. 2.3c-d and e-f). Both displacement field and energy flux amplitudes are two to three

orders of magnitude smaller in the elastic plate (for $b_1 < x_2 < b$) than in the poroelastic plate. The aluminum coating is almost acting as a rigid boundary condition, as also testified by the fact that the x_1 components of the displacement field and energy fluxes are almost symmetric with respect to the axis $x_2 = b_1/2$, while their x_2 components are almost antisymmetric along the x_2 axis. Note that the fluid phase Poynting vectors depicted in Fig. 2.3f are of equal modulus but opposite sign, symmetric with respect to the axis $x_2 = b_1/2$, and more than a hundred times smaller than total energy flux in amplitude. The energy of these modes are thus mainly localized in the solid phase of the poroelastic plate.

To go a step further, we evaluated the energy transport velocity $\overline{V}_e = \overline{P}_t / \overline{U}$ as the ratio between the average energy flux in each layer $\overline{P}_t$ over the mean total energy $\overline{U}$. The energy transport velocity of the mode highlighted in orange in Fig. 2.3a-b is compared to the group velocity $v_g = \partial\omega / \partial \text{Re}(k_1)$ ¹²⁹ in Fig. 2.3g. It is numerically computed by doing finite differences on a branch of the dispersion relation. Both velocities match for frequency ranges where the imaginary part of k_1 is low, i.e., for weak attenuation modes^{130–133}, but are different for strong attenuation. In particular, the group velocity diverges around the cut-on frequency. This is the reason why the energy transport velocity is usually preferred to study absorbing structures. The energy transport velocity $\overline{V}_e$ of the mode highlighted in green Fig. 2.3a-b is depicted in Fig. 2.3h. This energy transport velocity is always positive, while k_1 crosses the $\text{Re}(k_1) = 0$ axis. When $\text{Re}(k_1) < 0$ the phase velocity is negative, while the $\overline{V}_e > 0$. This branch (and the symmetric one) has also a negative group velocity as also testified by the fact $\text{Re}(k_1) < 0$ while $\text{Im}(k_1) > 0$ in the shaded area in Fig. 2.3h. This branch has also an exponentially growing amplitude.

2.4.4. Experimental validation

The experimental set-up is depicted in Fig. 2.4a. The sample is the two-layer structure considered in the previous subsection. The aluminum plate is glued on top of the poroelastic layer. The latter layer is glued on a 3 cm-thick aluminium plate, which mimic the rigid backing. The three elements are 45 cm wide and 85 cm long. The surrounding medium and the saturating fluid is air, at regular pressure and temperature conditions. The sample is positioned perpendicularly to the ground on its longest side as depicted in Fig. 2.4a to ease the mode excitation in the poroelastic layer. The excitation is provided by a shaker (Brüel and Kjaer type 4810), which is rigidly attached to the sample with a threaded steel rod fixed to the shaker on one side and glued on a 1mm thick aluminum plate of width 10 cm and height 1.5 cm. This plate is cut at the edge opposite to the threaded steel rod and glued to the porous sample, creating a line source at the edge of the sample. The resonance of this part was measured to be 4500 Hz, which is the upper limit of the measured frequency range. The excitation signal is a swept-sine function with 400 points ranging from 1 kHz to 3 kHz. The general layout of the experiments is similar to that used originally in¹³⁴. The key difference in our case is that the excitation is operated directly to the poroelastic layer and not to the top of the aluminum plate, because the modes that corresponds to propagating solutions in the poroelastic layer are encapsulated in between two almost rigid plates (see Section 2.4.3). The normal displacement is measured with a laser Doppler vibrometer (Polytec VibroFlex Neo) on a line of length $L = 40$ cm, along the side of the thickness of the porous layer. These measurements are averaged 100 times. The excitation signal and the measured field are interfaced through a computer via a Zürich Instruments acquisition card.

The SLaTCoW method is used to analyze the space-frequency displacement measurements and recover the complex wavenumber-real frequency dispersion relation. This method relies on the Laplace spatial transform of these measurements, which gives access to the real and imaginary components of the complex wavenumber. The difference between this Laplace transform and a correctly chosen ansatz function is finally minimized for each frequency, to recover the dispersion relation. More details on this method are given in Appendix 2.6.E as well as in the original article¹³⁴. The complex wavenumber-real frequency dispersion relation in Fig. 2.4b-c depicts the wavenumbers

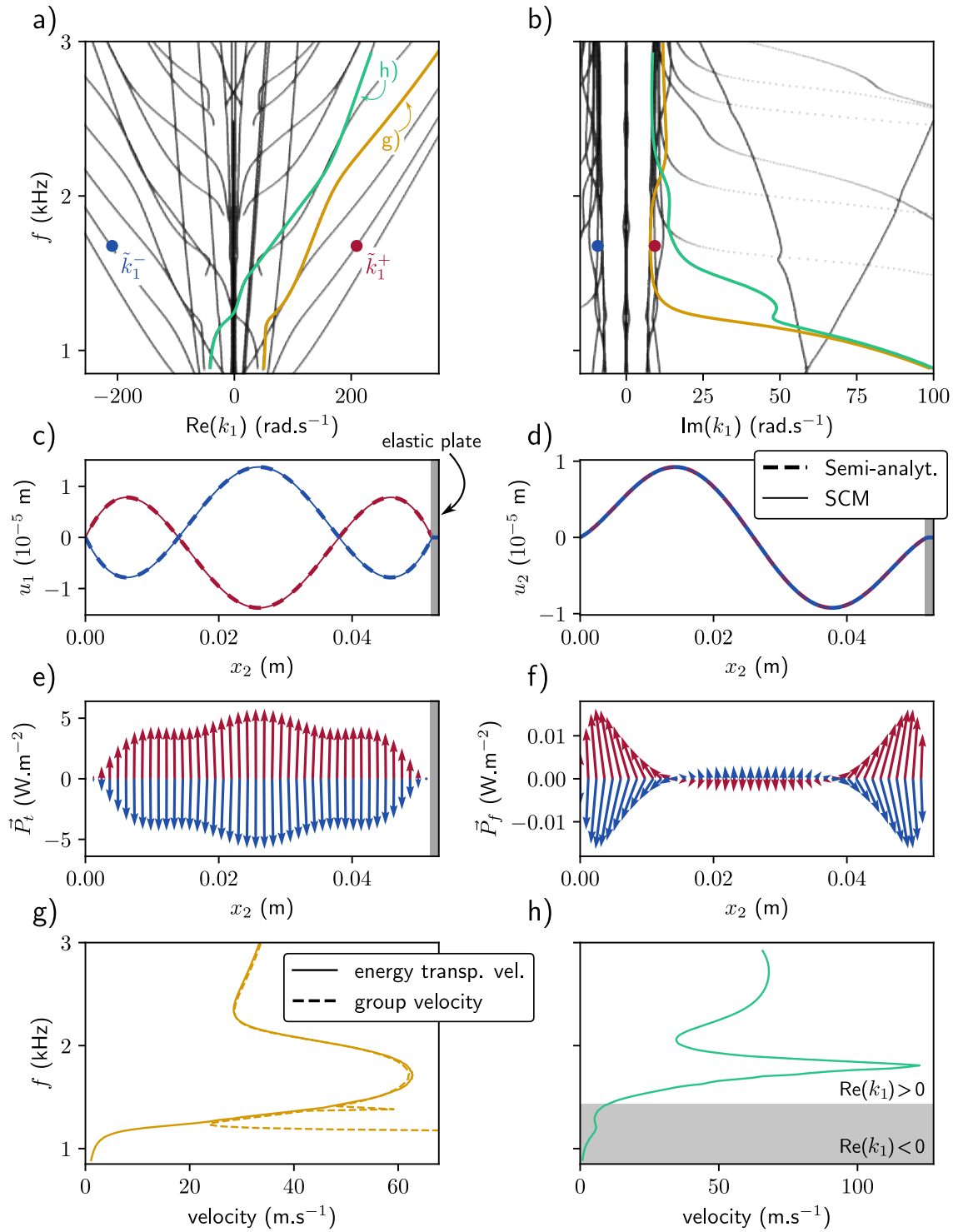


Fig. 2.3: a) and b) represent the real and imaginary parts of the dispersion relation of the two-layer structure. Two branches are highlighted in yellow and green, as well as two specific solutions, $\tilde{k}_1^-$ (blue) and $\tilde{k}_1^+$ (red), symmetric with respect to the $\text{Re}(k_1) = 0$ axis. Displacement fields in the x_1 c) and x_2 direction d) associated with the $\tilde{k}_1^+$ and $\tilde{k}_1^-$ solutions displayed. Total Poynting e) vector and fluid Poynting vector f) with the length of the arrow indicating the amplitude and angle indicating the direction of the vector field. g) Group and energy transport velocities for the yellow branch. h) Energy transport velocity for the green branch. The gray area represents the frequency range over which this velocity is negative.

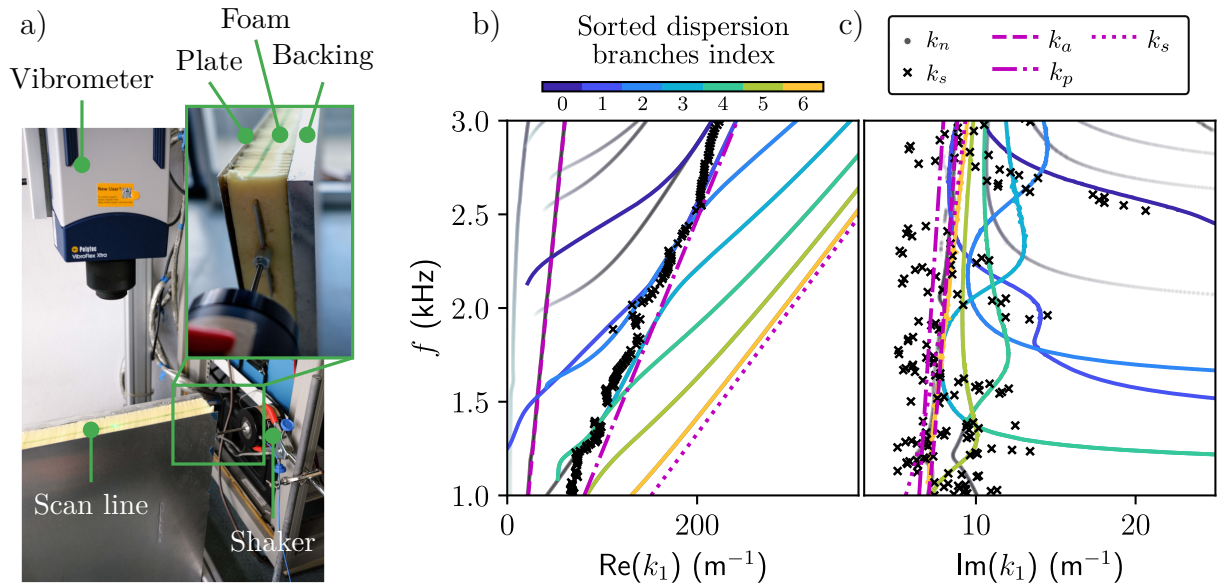


Fig. 2.4: a) Experimental set-up used for the measurement of the normal displacement field of the two-layer structure, with a close-up of the sample excitation. b) Real and c) imaginary parts of the dispersion relations recovered experimentally from the SLATCoW method k_s (black crosses) and calculated from the SCM k_s (colored lines and black-to-grey dots) and the bulk wavenumbers in the poroelastic material (magenta lines).

recovered experimentally k_s and simulated k_n . Four modes were sought during the minimization with the SLATCoW method: three of them correspond to the three bulk waves due to excitation and the fourth is depicted. The experimentally recovered mode jumps from one branch to the other because of the large modal density. The cut-on frequencies of the modes are clearly visible, especially on the imaginary part of the wavenumber. The lower real wavenumber parts of the numerical dispersion relation are not retrieved with the experimental results. This is because they correspond to a high attenuation of the mode, a part that is inherently difficult to measure in this experimental conditions. Generally, the recovered branches are those around the bulk solid compression wavenumber, corresponding to the wave that we most probably excite in the structure. The difficulties in efficiently exciting all the guided waves supported by this configuration largely explain the missing curves in the dispersion relation. The location and orientation of the excitation line limit possible symmetries and the generation of potential shear waves. This choice was made to avoid the main excitation of guided waves in the elastic plate when excitation and displacement measurement are performed only on this element. In addition, the modes mainly supported by the fluid phase should also be better excited by a loudspeaker and therefore better measured by a microphone. However, we have chosen to focus on the modes mainly supported by the solid phase. Nevertheless, the agreement between experimental results and simulation is quite good, thus validating the method.

2.5 | Conclusion

A spectral collocation method (SCM) is proposed to calculate dispersion relations for guided acoustic wave propagation in multilayer structures. The formalism is general enough to be able to consider arbitrary arrangements of (visco-) elastic, poroelastic, and (viscothermal-) fluid layers. Outgoing fluid radiation, leading to Leaky modes, can be accounted for. The fields in each layer is approximated by Chebyshev polynomials and discretized on the corresponding nodes. Interface

conditions are implemented to couple layers together. The solution provides the full complex wavenumber-real frequency dispersion relation of the considered structure. In addition, the physical fields in each layer are directly evaluated since they are the eigenvectors associated with the eigenvalue problem giving the dispersion relation. The dispersion diagram calculated with the present SCM is validated against experimental results and that calculated with a root-finding method (Müller method) in a simple two-layer structure consisting of a rigidly backed poroelastic layer covered by a thin aluminum plate radiating in a fluid half-space. The semi-analytical approach underlying the root-finding method is taken a step further to solve for the wave amplitudes as well, providing a reference for the mode shapes calculated with the SCM. The dispersion properties in such structures, as well as the direction of the energy density in each phase of the poroelastic layers using a decoupled expression for the Poynting vectors is analyzed. Results on energy transport velocity are also reported.

This method paves the way of complex wavenumber-real frequency dispersion relation calculation and analysis of more complex structures, like layers with embedded periodic inclusions (elastic or resonant), i.e. metaporolastic surfaces.

2.6 | Appendices

2.6.A. Expression of the Biot elastic coefficients and effective parameters in poroelastic materials

The complex and frequency dependent density and bulk modulus of the fluid phase, which accounts respectively for the viscous and thermal losses are^{112,113} as,

$$\begin{aligned}\rho_{eq}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - i \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda} \right)^2} \right), \\ K_{eq} &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) / \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \text{Pr} \omega} \sqrt{1 - i \text{Pr} \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1},\end{aligned}\quad (2.28)$$

where ϕ is the porosity, α_∞ is the tortuosity, Λ and Λ' are respectively the viscous and thermal characteristic lengths, R_f is the flow resistivity, μ_f is the dynamic viscosity, κ is the heat capacity ratio, ρ_f the density of the saturating fluid, and P_0 is the ambient pressure.

The Biot elastic coefficients, P , Q , and R , are more commonly expressed in terms of the frame, K_b , effective fluid, K_{eq} , and shear, N_s , moduli when saturated by a light fluid as

$$P = K_b + \frac{4}{3} N_s + (1 - \phi)^2 \frac{K_{eq}}{\phi}; \quad Q = K_f (1 - \phi), \quad R = \phi K_{eq}, \quad (2.29)$$

while the apparent densities are

$$\rho_{22} = \phi^2 \rho_{eq}, \quad \rho_{12} = \phi \rho_f - \rho_{22}, \quad \rho_{11} = (1 - \phi) \rho_s - \rho_{12}, \quad (2.30)$$

with ρ_s the density of the solid phase. The coefficients in Eq. (2.13) are thus

$$\hat{A} = K_b - \frac{2}{3} N_s + \frac{K_{eq}}{\phi} ((1 + (1 - \phi))), \quad \hat{\rho} = \rho_1 + \phi \rho_f - 2 \phi^2 \rho_{eq} - \frac{\rho_f^2}{\rho_{eq}} \quad \beta = \frac{\rho_f}{\rho_{eq}} - 2 \phi - 1, \quad (2.31)$$

where $\rho_1 = \phi \rho_f + (1 - \phi) \rho_s$ is the apparent density of the frame. Finally, the frame bulk modulus and shear modulus are linked by the Poisson ratio ν with $K_b = 2 N_s (1 + \nu) / 3 (1 - 2 \nu)$.

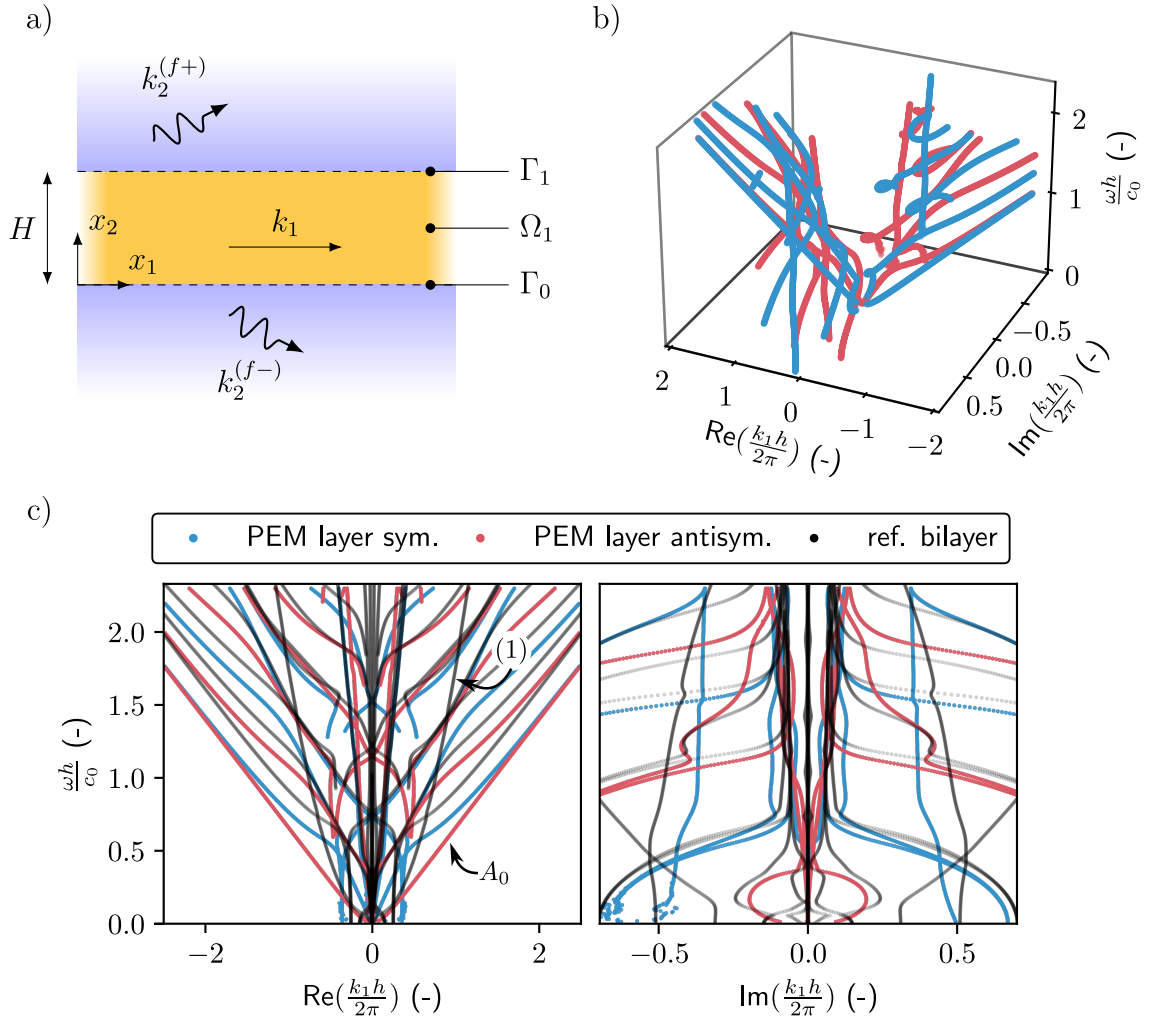


Fig. 2.5: a) Sketch of the single layer poroelastic structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. The branches of the symmetric and antisymmetric modes are represented by blue and red curves respectively. The dispersion relation of the two-layer elasto-poroelastic system studied in Section 2.4 are reminded with black lines.

2.6.B. Application to a single poroelastic layer surrounded by two fluid half-spaces

The dispersion relation of guided waves in a single layer of poroelastic material surrounded by two identical fluid half-spaces is considered (See Fig. 2.5a). The poroelastic material properties are those considered in Section 2.4.2 and the layer thickness is $2h^{(1)} = 104$ mm, i.e. twice the thickness of the poroelastic layer considered in the two-layer system.

The discretized equations of motion is that provided in Eq.(2.16). $M^{(1)} = 11$ collocation points are considered. Only the boundary conditions at the lower, i.e. Γ_0 at $x_2 = 0$, and upper, i.e. Γ_1 at $x_2 = h$, interfaces are modified. They read as¹²⁸

$$\begin{aligned} \left(1 + \phi + \phi \frac{\rho_{12}}{\rho_{22}}\right) u_2^{(1)} + \frac{\phi^2}{\rho_{22} \omega^2} \partial_2 p^{(1)} &= u_2^{(0\pm)}, \quad p^{(1)} = p^{(0\pm)}, \\ \hat{\sigma}_{22}^s &= \left(1 - \phi \left(1 + \frac{Q}{R}\right)\right) p^{(0\pm)}, \quad \hat{\sigma}_{12}^s = 0. \end{aligned} \quad (2.32)$$

The discretized form of these boundary conditions is

$$\underline{\underline{C}}_1 = \begin{pmatrix} \frac{\phi^2}{\rho_{22}} \underline{\underline{D}}_2^+ & \underline{0} & \omega^2 \left(1 + \phi \left(1 + \frac{\rho_{12}}{\rho_{22}} \right) \right) \underline{I}^+ & \frac{ik_2^{(f)}}{\rho_f} \\ \underline{I}^+ & \underline{0} & \underline{0} & -1 \\ \underline{0} & ik_1(\hat{P} - 2N_s) \underline{I}^+ & \hat{P} \underline{\underline{D}}_2^+ & 1 - \phi \left(1 + \frac{Q}{R} \right) \\ \underline{0} & N_s \underline{\underline{D}}_2^+ & ik_1 N_s \underline{I}^+ & 0 \end{pmatrix}. \quad (2.33)$$

These matrices are then arranged in the same manner as Eq.(2.10) and the wavenumbers are calculated. Fig. 2.5b-c depicts the results.

In a similar way as Lamb modes, the different modes can be sorted in symmetric and antisymmetric modes. This is done by comparing the sign of the radiated amplitudes at each edge of the layer. Results are shown in Fig. 2.5b. All modes necessarily have an imaginary part, both because of the radiation condition, i.e. leakage, and because of viscothermal dissipation as can be noticed in Fig. 2.5c.

2.6.C. Evaluation of the physical fields from the SCM solutions

The discrete sets of eigenvectors resulting from the SCM are the spatial Laplace transforms of some physical fields at collocation points, i.e. the locations where the discrete equations of motion (residue) are exactly satisfied. These discrete fields are interpolated into a continuous form using

$$s^{(n)}(x_2) = \left[\psi_m^{-1}(\underline{\xi}) \underline{s}_m^{(n)}(\underline{\xi}) \right] \psi(\xi^{(n)}) \quad (2.34)$$

where the terms in brackets correspond to the coefficients $\alpha_m^{(n)}$ given in Eq. (2.8). The Chebyshev polynomials ψ is the basis that maps from the discrete grid $\underline{\xi}$ back to the physical space.

The stress tensor components depend on the direct expressions of the fields and their first order derivative in x_2 , denoted with a prime symbol in the following. The various fields, differentiated up to any order, can be calculated by replacing ψ by its derivative in Eq. (2.34). Explicit expressions are given in the following,

- in the elastic layer, the spatial Laplace transforms of stress tensor components are computed from $\tilde{\mathbf{u}}$, with

$$\begin{aligned} \tilde{\sigma}_{11} &= (\lambda + 2\mu) ik_1 \tilde{\mathbf{u}}_1 + \lambda \tilde{\mathbf{u}}_2', & \tilde{\sigma}_{12} &= \mu (\tilde{\mathbf{u}}_1' + ik_1 \tilde{\mathbf{u}}_2), \\ \tilde{\sigma}_{22} &= \lambda ik_1 \tilde{\mathbf{u}}_1 + (\lambda + 2\mu) \tilde{\mathbf{u}}_2'. \end{aligned} \quad (2.35)$$

- in the poroelastic layer, the full solid stress tensor is expressed as $\tilde{\sigma}_s = \hat{\sigma}_s - \phi (Q/R) \tilde{\mathbf{p}}$, with $\hat{\sigma}_s$ the spatial Laplace transform of the *in vacuo* solid stress tensor introduced in Section 2.4. The components of this full solid stress tensor are,

$$\begin{aligned} \tilde{\sigma}_{s,11} &= P ik_1 \tilde{\mathbf{u}}_{s,1} + \hat{A} \tilde{\mathbf{u}}_{s,2}' - \phi \frac{Q}{R} \tilde{\mathbf{p}}, & \tilde{\sigma}_{s,12} &= N_s (\tilde{\mathbf{u}}_{s,1}' + ik_1 \tilde{\mathbf{u}}_{s,2}), \\ \tilde{\sigma}_{s,22} &= \hat{A} ik_1 \tilde{\mathbf{u}}_{s,2} + P \tilde{\mathbf{u}}_{s,2}' - \phi \frac{Q}{R} \tilde{\mathbf{p}}. \end{aligned} \quad (2.36)$$

The Laplace transform of the fluid stress tensor is calculated as

$$\tilde{\sigma}_f = -\phi \tilde{\mathbf{p}} \delta_{ij}, \quad (2.37)$$

and the fluid displacement components are

$$\tilde{\mathbf{u}}_{f,1} = \frac{\phi}{\rho_{22} \omega^2} ik_1 \tilde{\mathbf{p}} - \frac{\rho_{12}}{\rho_{22}} \tilde{\mathbf{u}}_{s,1}, \quad \tilde{\mathbf{u}}_{f,2} = \frac{\phi}{\rho_{22} \omega^2} \tilde{\mathbf{p}}' - \frac{\rho_{12}}{\rho_{22}} \tilde{\mathbf{u}}_{s,2}. \quad (2.38)$$

All physical fields $\theta(x)$ can finally be calculated by the inverse spatial Laplace transform with $\theta(x) = \tilde{\theta}(k_1, x_2)e^{ik_1x_1}$.

2.6.D. Energy flux in the multilayer structure - Poynting theorem in a poroelastic medium

Energy conservation is commonly conducted in the time domain. Let us consider a volum Ω , bounded by $\partial\Omega$, with $\mathbf{n}$ the outgoing normal vector. The energy balance or Poynting theorem in a poroelastic materials reads as^{88,135}

$$\begin{aligned} - \int_{\partial\Omega} (-\sigma_s \cdot \mathbf{v}_s^*) \cdot \mathbf{n} + (-\sigma_f \cdot \mathbf{v}_f^*) \cdot \mathbf{n} dS &= \frac{1}{2} \int_{\Omega} \rho_1 \partial_t |\mathbf{v}_s|^2 - \rho'_{12} \partial_t (\mathbf{v}_f - \mathbf{v}_s)^2 + \rho_2 \partial_t |\mathbf{v}_f|^2 d\Omega \\ &+ \int_{\Omega} \sigma_s : \partial_t \epsilon_s^* + \sigma_f : \partial_t \epsilon_f^* - b (\mathbf{v}_s - \mathbf{v}_f)^2 d\Omega, \end{aligned} \quad (2.39)$$

where $\mathbf{v}_s = \partial_t \mathbf{u}_s$ and $\mathbf{v}_f = \partial_t \mathbf{u}_f$ are the velocity fields, σ_s and σ_f are the stress tensors, $\epsilon_s = 1/2 (\nabla \mathbf{u}_s + \nabla^T \mathbf{u}_s)$ and $\epsilon_f = 1/2 (\nabla \mathbf{u}_f + \nabla^T \mathbf{u}_f)$ are the strain tensors in the solid frame and in the fluid phase respectively, and $\star$ denotes the complex conjugate. Note that a similar expression is provided in¹³³, where the alternative 1962 Biot formulation is used.

The left-hand side contains the energy density flux $\mathbf{P}_s = -\sigma_s \cdot \mathbf{v}_s^*$ and $\mathbf{P}_f = -\sigma_f \cdot \mathbf{v}_f^*$ in the elastic frame and fluid phase. The right-hand side contains the kinetic energy K ¹¹ and the internal forces $P_{\text{int}} = \partial_t W + P_{\text{dis}}$. The latter is the sum of the time derivative of the strain energy W and the dissipated power P_{dis} by the elastic damping and the viscous and thermal losses¹³⁵. In the frequency domain, the kinetic and strain energies read as

$$\begin{aligned} K &= \frac{1}{2} \rho_1 \partial_t |\mathbf{v}_s|^2 - \frac{1}{2} \rho'_{12} \partial_t (\mathbf{v}_f - \mathbf{v}_s)^2 + \frac{1}{2} \rho_2 \partial_t |\mathbf{v}_f|^2, \\ W &= \frac{1}{2} \hat{\sigma}_s : \epsilon_s^* + \frac{1}{2} R \nabla \cdot (Q \mathbf{u}_s + R \mathbf{u}_f), \end{aligned} \quad (2.40)$$

Finally, the time average total Poynting vector is $\langle \mathbf{P}_t \rangle = \langle \mathbf{P}_s \rangle + \langle \mathbf{P}_f \rangle$, with $\langle \mathbf{P}_s \rangle = -\text{Re}(\sigma_s \cdot \mathbf{v}_s^*)/2$ and $\langle \mathbf{P}_f \rangle = -\text{Re}(\sigma_f \cdot \mathbf{v}_f^*)/2$ and the total energy density is $\langle U \rangle = \langle K \rangle + \langle W \rangle = \text{Re}(K)/2 + \text{Re}(W)/2$. Note that a similar procedure can be followed for isotropic elastic or fluid medium¹³⁶. In particular, the elastic Poynting vector is $\langle \mathbf{P} \rangle = \text{Re}(\sigma \cdot \mathbf{v}^*)/2$, where σ is the stress tensor and $\mathbf{v}$ is the velocity in the elastic medium.

The energy transport velocity V_e is evaluated from the latter expressions

$$V_e = \frac{\langle \mathbf{P}_t \rangle \cdot \mathbf{n}}{\langle U \rangle}. \quad (2.41)$$

When a multi-layer configuration is considered, the latter velocity is averaged over the structure thickness $\bar{V}_e = \frac{\bar{P}_t}{\bar{U}}$, with the average Poynting vector and the average total energy defined as

$$\bar{P}_t = \sum_n \frac{1}{h_n} \int_{h_{n-1}}^{h_n} \langle \mathbf{P}_t^{(n)} \rangle \cdot \mathbf{e}_{x_2} dx_2, \quad \bar{U} = \sum_n \frac{1}{h_n} \int_{h_{n-1}}^{h_n} \langle U^{(n)} \rangle dx_2. \quad (2.42)$$

2.6.E. Reminder of the SLATCoW method for obtaining complex-valued wavenumbers.

The key point of the SLATCoW method is the Laplace transform, $\mathfrak{L}_{\text{mes}}(z)$ with $z = z_r + iz_i$ the Laplace space coordinate, of the data measured in space at a fixed frequency ω . In doing so, we

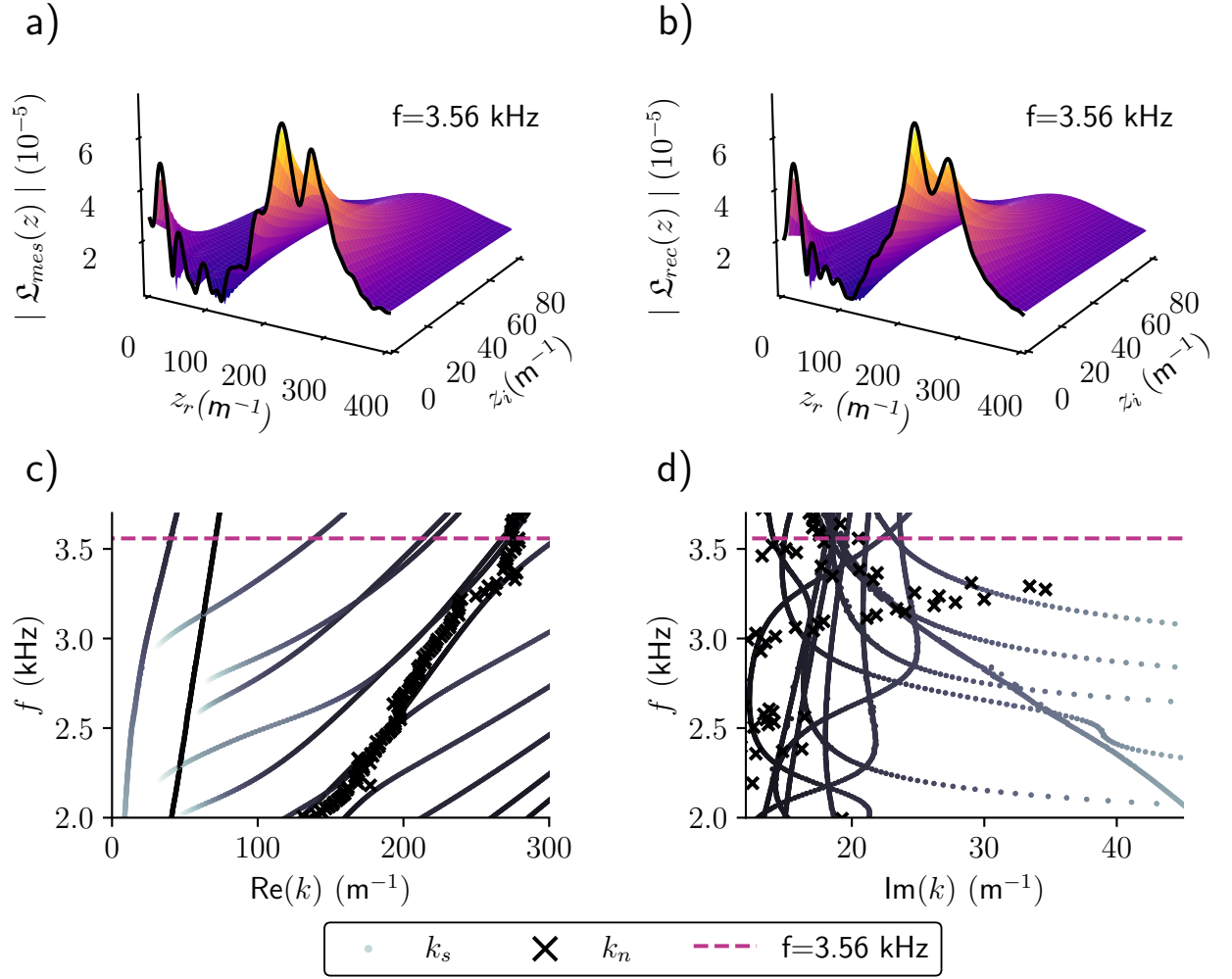


Fig. 2.6: Description of the procedure of the SLatCoW method: a) Spatial Laplace transform from the experimental data, b) Spatial Laplace transform of the reconstructed wave fields. Real c) and imaginary d) parts of the experimentally retrieved wavenumbers (crosses) compared to the SCM (black-to-grey dots). The frequency at which the Laplace transforms are represented corresponds to the magenta dashed line in the graph.

move from the spatial domain to the complex wavenumber domain. The difference between $\mathfrak{L}_{\text{mes}}(z)$ and the spatial Laplace transform of the displacement ansatz along the same line, $\mathfrak{L}_{\text{rec}}(z)$, is then minimized in the least square sense to recover both the complex amplitude (magnitude and phase) and the complex wavenumber of each mode assumed to be part of the displacement ansatz.

The experimental Laplace spatial transform $\mathfrak{L}_{\text{mes}}(z) = \int_0^L r(x)e^{-izx}dx$ is computed at each frequency ω from the transfer function $r(x)$ between the excitation and the normal displacement measured along a line $x \in [0, L]$, with $L = 0.3$ m and spacing $d_x = 0.001$ m. The displacement ansatz is assumed to be of the form $w(x) = \sum_{\beta} A^{\beta} e^{ik^{\beta}x}$, with β the number of guided-waves, i.e.,

the number of modes, and where $A^{\beta} = |A^{\beta}|e^{i\phi^{\beta}}$ is the complex amplitude and k^{β} is the complex wavenumber of the β -th mode.. In practice, β is the smallest integer that gives a good fit between the two Laplace transforms and therefore depends on the frequency under consideration. Its spatial Laplace transform reads as $\mathfrak{L}_{\text{rec}}(z) = \sum_{\beta} A^{\beta} L e^{i(k^{\beta}-z)\frac{L}{2}} \text{sinc}\left((k^{\beta}-z)\frac{L}{2}\right)$.

The cost function $\mathfrak{C} = (\|\mathfrak{L}_{\text{rec}}(z) - \mathfrak{L}_{\text{mes}}(z)\|) / \|\mathfrak{L}_{\text{mes}}(z)\|$ is finally minimized using the Matlab function *fminsearchbnd* for $\{|A^{\beta}|, \phi^{\beta}, \text{Re}(k^{\beta}), \text{Im}(k^{\beta})\}, \forall \beta$. The initial guess are the recovered

values of the minimization parameters at the previous frequency, with the exception of the first frequency. The method is graphically exposed in Figure 2.6. Figures 2.6 c) and d) depict the complex dispersion relation as already depicted in Fig. 2.4. Figure 2.6a) depicts the spatial Laplace transform of the experimental data and Figure 2.6b) depicts $\mathfrak{L}_{\text{rec}}(z)$ with the parameters derived from minimization of the cost function at the frequency $f = 3.56$ kHz. This frequency is highlighted by the purple dashed line in Figures 2.6 c) and d). The two Laplace transforms match almost perfectly.

Appendices

Contents

A	Benchmark of metaporous configurations	41
B	Implementing SAFE to compute the dispersion relation of Lamb modes	41
	B.1 Formalism	41
	B.2 Results and convergence comparison	42
C	Details on the root-finding algorithm	44
	C.1 The Müller method	44
	C.2 A complete routine for the automated solving of dispersion relations . . .	44
D	Overview of activities and scientific works during the thesis	46
	D.1 Publications	46
	D.2 Conferences	46
	D.3 Workshops and training schools	46
	D.4 Teaching activities	46
E	Extended abstract in french	47

A | Benchmark of metaporous configurations

TODO: 7 ou 8 configurations du benchmark avec coefficient d'absorption COMSOL / py-PLANES (+ MST quand possible ?) avec des schémas

B | Implementing SAFE to compute the dispersion relation of Lamb modes

As discussed in Section. 2.2, the SAFE method is an alternative to SCM for the numerical computation of dispersion relation of guided waves in layers. They rely on a similar formalism, that is a 1D discretization across the thickness of each layer and an assumption of guided harmonic waves along the longitudinal direction. An analogous method to that presented in Chapter 2 could be developed using SAFE, the key difference being the location of nodes in the layer and the interpolation function used for discretization. There is a large bibliography dealing with these topics and applying this method to similar problem to ours.

The general theory of SAFE is explained in great details in Chapter 9 of Rose's textbook¹⁰² hence The present method is inspired from. Examples of SAFE application to dispersion curve calculation are illustrated by Predoi *et al.*¹³⁷ where a numerical model was implemented to investigate periodic structure of infinite width with the addition of periodic boundary conditions allowing for the computation of guided wave modes. A similar approach was generalized for systems with arbitrary cross-sections and a coupling with infinite solid media by using the so-called absorbing region¹³⁸. Other description of the surrounding medium where leakage happens have also been investigated by using a PML¹³⁹. This has been done by Mazzotti *et al.* using BEM^{140,141}. A local coupling of the surrounding space was done in Ref.¹⁴² by linearizing a non-linear GEP, a similar approach to that of Refs.^{P1,109}.

In this appendix, we propose to quickly re-derive a SAFE method inspired from these previous papers to benchmark the two numerical methods regarding their respective performances. We will use high-order Lobatto function as interpolating function of the finite-elements core of the method. The system studied here is the simple of Lamb modes: an elastic layer of infinite extent in the x_1 direction has two free edges at $x_2 = 0$ and $x_2 = b$, where a no-traction boundary condition is applied.

B.1. Formalism

Let us start with the general variational form for the elasticity equation

$$-\rho\omega^2 \int_{\Omega} u \delta u \, d\Omega + \int_{\Omega} \sigma(u) : \epsilon(\delta u) \, d\Omega = \int_{\partial\Omega} \sigma(\delta u) \cdot \mathbf{n} \, d\Gamma, \quad \forall \delta u, \quad (\text{B.1})$$

where the displacement field is

$$\mathbf{u}(\mathbf{x}) = \mathbf{u}(x_2) e^{ik_1 x_1} e^{-i\omega t}, \quad (\text{B.2})$$

and the double-dot product is $\sigma(u) : \epsilon(\delta u^*) = \sigma_{ij}(u) \epsilon_{ij}(\delta u^*)$. σ designates the stress tensor while ϵ is the strain tensor. Developing this part of the volume term gives the expression,

$$\begin{aligned} \sigma_{11}\epsilon_{11} &= k_1^2(\lambda + 2\mu)u_1\delta u_1 + -ik_1\lambda\nabla u_2\delta u_1 \\ \sigma_{12}\epsilon_{12} &= \frac{\mu}{2} \left(-ik_1\nabla u_1\delta u_2 + \nabla u_1\delta\nabla u_1 + k_1^2u_2\delta u_2 + ik_1u_2\nabla\delta u_1 \right) \\ \sigma_{22}\epsilon_{22} &= ik_1\lambda u_1\nabla\delta u_2 + (\lambda + 2\mu)\nabla u_2\nabla\delta u_2. \end{aligned} \quad (\text{B.3})$$

Putting these terms back into Eq. B.1 yields,

$$\begin{aligned} \int_{\Omega} -\rho\omega^2 (u_1\delta u_1 + u_2\delta u_2) + k_1^2(\lambda + 2\mu)u_1\delta u_1 - ik_1\lambda\nabla u_2\delta u_1 - ik_1\mu\nabla u_1\delta u_2 + \mu\nabla u_1\delta\nabla u_1 \\ + \mu k_1^2 u_2\delta u_2 + ik_1\mu u_2\nabla\delta u_1 + ik_1\lambda u_1\nabla\delta u_2 + (\lambda + 2\mu)\nabla u_2\nabla\delta u_2 d\Omega = 0 \quad \forall \delta u \end{aligned} \quad (\text{B.4})$$

where ∇u_i denotes differentiation of the i -th component of the displacement along x_2 . Note that the surface term initially expressed vanishes from the no-traction boundary condition.

The displacement fields can be discretized as

$$\mathbf{u}(\mathbf{x}) = \mathbf{N}(\xi)\mathbf{U}e^{ik_1x_1}e^{-i\omega t}, \quad (\text{B.5})$$

with $\mathbf{N}$ containing the Lobatto shape functions, and $\mathbf{U}$ is the vector of displacement amplitudes. Following the same discretization, the test function thus becomes $\forall \delta \mathbf{U}$. By rearranging the terms of Eq. B.4 and writing the weak form for one element,

$$\begin{aligned} \int_{\Omega_e} k_1^2 \left(\mu \delta \mathbf{U}_2^T \mathbf{M}^{(e)} \mathbf{U}_2 + (\lambda + 2\mu) \delta \mathbf{U}_1^T \mathbf{M}^{(e)} \mathbf{U}_1 \right) + ik_1 \left(\mu \delta \mathbf{U}_2^T \mathbf{C}_1^{(e)} \mathbf{U} - \mu \delta \mathbf{U}_1^T \mathbf{C}_2^{(e)} \mathbf{U}_2 \right. \\ \left. + \lambda \delta \mathbf{U}_1^T \mathbf{C}_1^{(e)} \mathbf{U}_2 - \lambda \delta \mathbf{U}_2^T \mathbf{C}_2^{(e)} \mathbf{U}_1 \right) + \mu \delta \mathbf{U}_1^T \mathbf{K}^{(e)} \mathbf{U}_1 + (\lambda + 2\mu) \delta \mathbf{U}_2^T \mathbf{K}^{(e)} \mathbf{U}_2 \\ - \rho\omega^2 \delta \mathbf{U}_1^T \mathbf{M}^{(e)} \mathbf{U}_1 - \rho\omega^2 \delta \mathbf{U}_2^T \mathbf{M}^{(e)} \mathbf{U}_2 d\Omega_e = 0 \end{aligned} \quad (\text{B.6})$$

where the elementary matrices are,

$$\begin{aligned} \mathbf{M}^{(e)} &= \frac{b}{2} \int_{-1}^1 \mathbf{N}^T \mathbf{N} d\xi, \quad \mathbf{K}^{(e)} = \frac{2}{b} \int_{-1}^1 \nabla \mathbf{N}^T \nabla \mathbf{N} d\xi, \\ \mathbf{C}_1^{(e)} &= \int_{-1}^1 \mathbf{N}^T \nabla \mathbf{N} d\xi, \quad \mathbf{C}_2^{(e)} = \int_{-1}^1 \nabla^T \mathbf{N}^T \mathbf{N} \mathbf{U} d\xi, \end{aligned} \quad (\text{B.7})$$

In Eq. B.6, the dependance on k_1 was highlighted and terms were sorted depending on their order. The elementary matrices should be assembled into the global matrices and then, following an analogous procedure to Chapter. 2, the general form for the problem becomes,

$$\left(k_1^2 \mathbf{A}_2 + ik_1 \mathbf{A}_1 + \mathbf{A}_0 - \omega^2 \mathbf{M} \right) \mathbf{U} = 0. \quad (\text{B.8})$$

It can easily be rewritten as a generalized eigenvalue problem,

$$\left[\begin{pmatrix} -\mathbf{A}_1 & \mathbf{A}_0 - \omega^2 \mathbf{M} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} - k_1 \begin{pmatrix} \mathbf{A}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \right] \begin{pmatrix} k_1 \mathbf{U} \\ \mathbf{U} \end{pmatrix} = 0, \quad (\text{B.9})$$

which solving is thus quite straightforward.

B.2. Results and convergence comparison

Analytical results are obtained by applying the root-finding approach using the Müller method presented in the paper. Solutions are sought from the classical Lamb modes equations, derived in many books such as Royer. Figure 2.7a) shows the results of the three methods for a given frequency range. They all match well. In order to compare the two numerical methods in play, a brief convergence study has been done using the analytical results from the root-finding method as a reference.

The convergence curve of the results given in Fig. 2.7b) is unsurprisingly in agreement with the literature. It is calculated at a frequency $f = 4$ kHz: the solutions along the dashed line in Fig 2.7a)

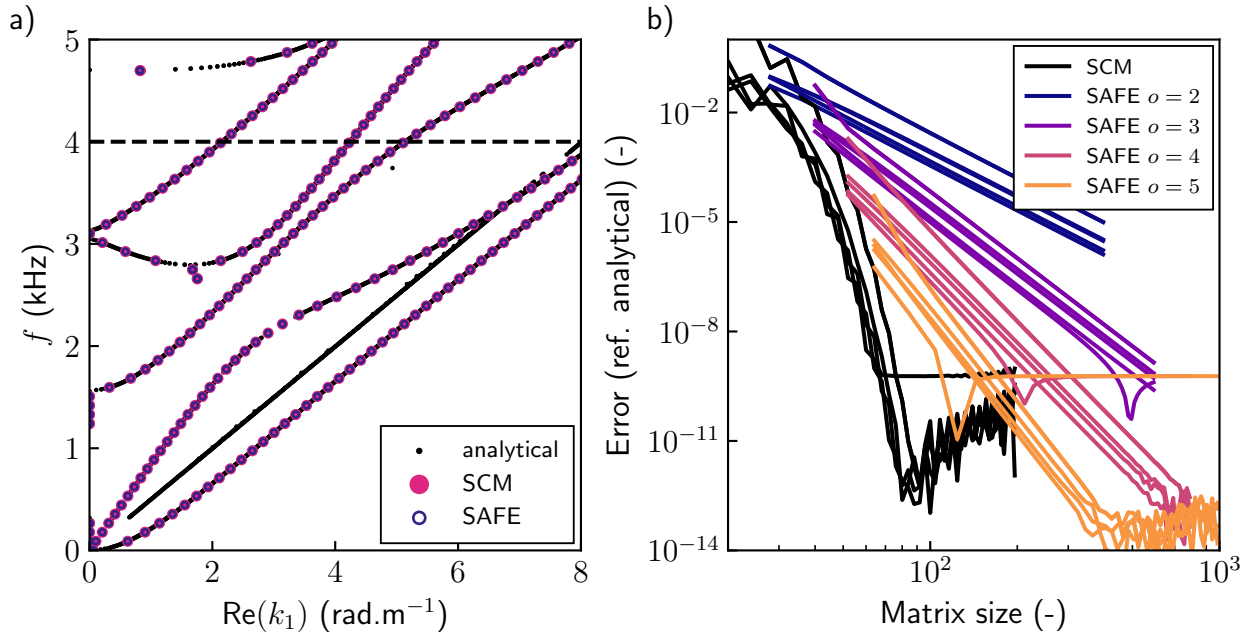


Fig. 2.7: a) Dispersion relation for the Lamb modes with the root-finding results, alongside the spectral collocation and semi-analytical finite elements solutions. b) Convergence curve of the 2 methods at a given frequency $f = 4$ kHz. The multiple lines depicts the convergence of each wavenumber.

are selected. The parameter n used for the discretization is shared between the two methods and corresponds to the number of elements taken for the discretization in each method. SAFE matrices have a size $2(2n + 1)$, while SCM matrices have a size $2n$. The discretization is swept from a number of nodes $n = 3$ to $n = 50$.

Matrix size for scm $4n$ while safe is $4(o \cdot n + 1)$ with o the order of Lobatto shape functions

The SCM convergence shows a steep slope and reaches a plateau around $n = 20$, while the SAFE curve follows the expected slope of a finite elements method that use second-order shape functions. When comparing at a given discretization order, the error from the SAFE is much lower than that of SCM by several orders of magnitude. In order to see comparable convergence, it would require to use higher-order shape functions or use spectral elements instead.

C | Details on the root-finding algorithm

C.1. The Müller method

equations, iterative scheme etc.

143

$$\begin{aligned}\Sigma_x &= \prod_{0 \leq i < j \leq 2} (x_j - x_i) = (x_0 - x_2)(x_1 - x_2)(x_0 - x_1) \\ a &= \frac{(x_1 - x_2)(f_0 - f_2) - (x_0 - x_2)(f_1 - f_2)}{\Sigma_x} \\ b &= \frac{(x_0 - x_2)^2(f_1 - f_2) - (x_1 - x_2)^2(f_0 - f_2)}{\Sigma_x} \\ c &= f_2\end{aligned}\tag{C.1}$$

$$f(x_3) = ax_3^2 + bx_3 + c\tag{C.2}$$

$$x_3 = x_2 - \frac{2c}{b + \text{sign}(b)\sqrt{b^2 - 4ac}}\tag{C.3}$$

$$\varepsilon_a = \left| \frac{x_3 - x_2}{x_3} \right|\tag{C.4}$$

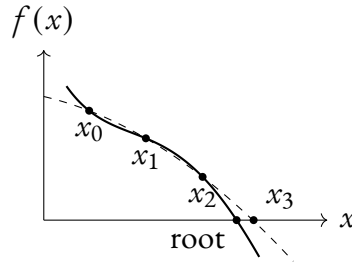


Fig. 2.8: iteration muller

C.2. A complete routine for the automated solving of dispersion relations

In order to find solutions for modes in the form of complex wavenumbers k_1 , we use the previously derived system from Eq. (??), that can also be expressed as $\mathbf{M}(k_1, \omega)\mathbf{S} = 0$. The modes are found by finding the discrete set of k_1 that satisfy the singularity of $\mathbf{M}$ using Eq. ??

In practice, these solutions are found using a root-finding procedure, our own implementation of the Müller method⁹⁵. It is based on a parabolic interpolation to iteratively finds complex wavenumber solutions k_1 given a specific frequency ω from three initial guesses. Therefore, care has to be put into the initialization of the method. Since the system $\mathbf{M}$ is constructed from a finite elements scheme, computations will be all the more costly that a large number of iterations need to run before convergence is reached. The convergence criterion checks that threshold of the variation of the solution is below ε . In this paper, this criterion is taken with a magnitude around $\varepsilon = 10^{-13}$. Thus, the initialization of the method is done by evaluating the local minimas of a coarse mapping of the function in the complex plane. The latter is only computed every few frequency steps to save

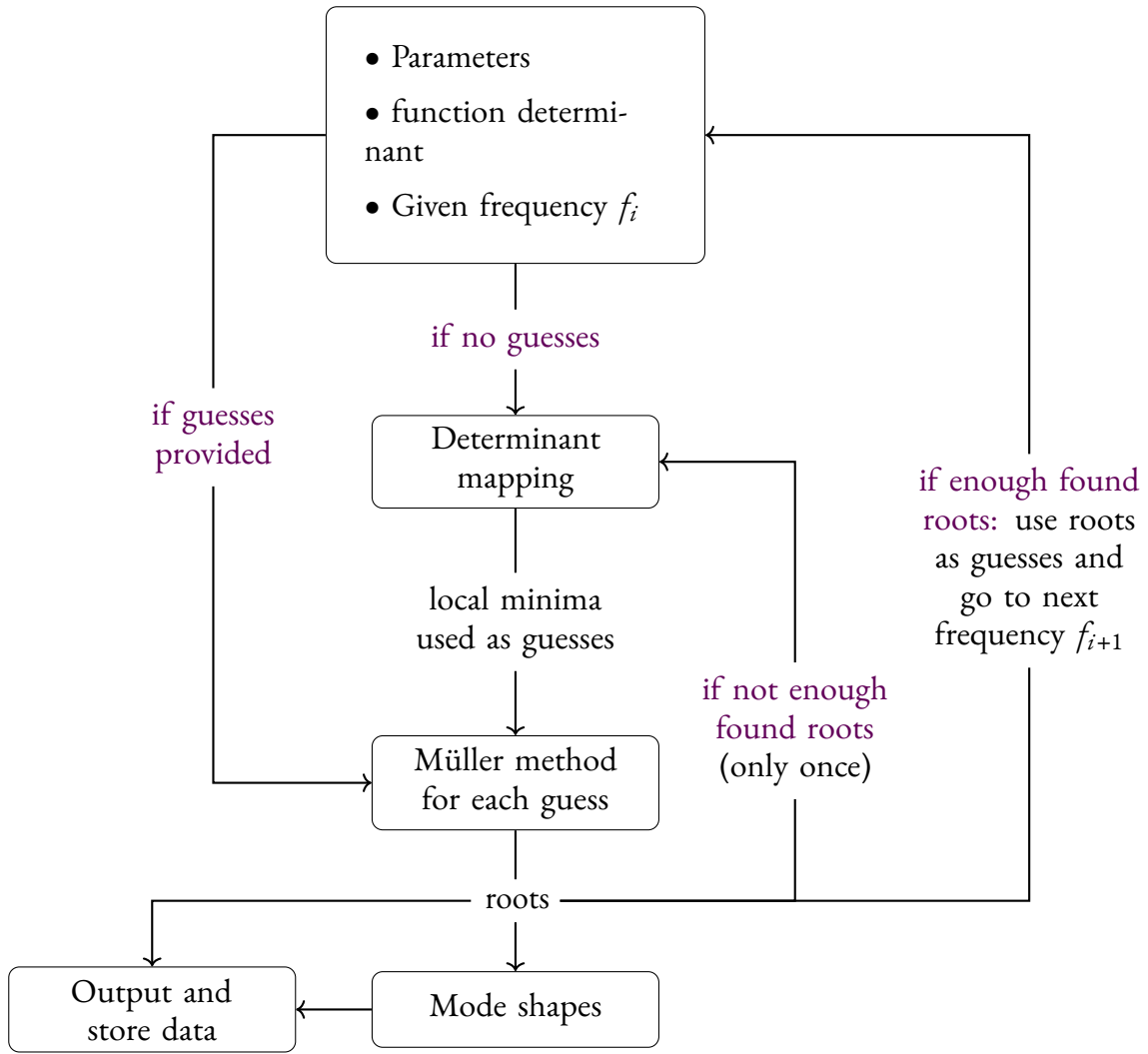


Fig. 2.9: Description of the routine

computation time. Additionally, the matrix D_{ii} from Eq. (??) does not depend on k_1 and can thus be only evaluated once per frequency step, thus saving valuable computing resources.

Once the set of solutions has been obtained, each of the associated mode shapes can also be retrieved. Given a specific root $k_{1,j}$ that satisfies $\det(\mathbf{M}) = 0$, there is $\mathbf{M}(k_{1,j})\mathbf{S} = 0$, where the vector $\mathbf{S} = \ker(\mathbf{M}(k_{1,j}))$ is the nullspace of the matrix. In practice, the nullspace is computed using singular value decomposition (SVD), $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\dagger$. The last right singular vectors, that is the last row of $\mathbf{V}^\dagger$ provide an orthonormal basis for the null space of $\mathbf{M}$, associated to the singular value $\sigma = \min(\mathbf{\Sigma})$. All state vectors among which S^t and S^b are obtained by selecting the appropriate rows of $\mathbf{V}^*$. By doing the whole procedure backwards, using Eq. (??) and then Eq. (??), we can retrieve the full degrees of freedom vector $\mathbf{p}$ (and normalized as $(p^i(x_1, x_2)/p^i(0, h))e^{ik_1x_1}$).

D | Overview of activities and scientific works during the thesis

D.1. Publications

- P1. M. Maréchal, A. Geslain, J.-P. Groby, V. Romero-García, and O. Dazel. “A General Spectral Collocation Method for Computing the Dispersion Relations of Guided Acoustic Waves in Multilayer Dissipative Structures”. *J. App. Phys.* 137.10 (2025), 104902.

a

D.2. Conferences

(ajouter titres des présentations)

- SAM 2022 Nimes, France, October 2022 (attendee)
- JJCAB Toulouse 2023, France, July 2023 (poster)
- SAPEM Sorrento, Italy, October 2023 (presenter)
- Séminaire de 2e année Centrale Nantes, France, April 2024 (presenter)
- SAM Aegina, Greece, May 2024 (presenter)
- INTERNOISE Nantes, France, August 2024 (attendee and volunteer)
- CFA 2025 Paris, France (presenter)

D.3. Workshops and training schools

- training school Acoustic Metamaterials Valencia Spain, UPV, November 2023
- lamb waves workshop

D.4. Teaching activities

E | Extended abstract in french

(en 10 à 15 k caractères)

References

1. M. Basner et al. "Auditory and Non-Auditory Effects of Noise on Health". *The Lancet* 383.9925 (2014), 1325–1332.
2. T. Münzel, F. P. Schmidt, S. Steven, J. Herzog, A. Daiber, and M. Sørensen. "Environmental Noise and the Cardiovascular System". *Journal of the American College of Cardiology* 71.6 (2018), 688–697.
3. O. Hahad et al. "Noise and Mental Health: Evidence, Mechanisms, and Consequences". *J Expo Sci Environ Epidemiol* 35.1 (2025), 16–23.
4. R. Sordello, O. Ratel, F. Flamerie De Lachapelle, C. Leger, A. Dambry, and S. Vanpeene. "Evidence of the Impact of Noise Pollution on Biodiversity: A Systematic Map". *Environ Evid* 9.1 (2020), 20.
5. C. Erbe, S. A. Marley, R. P. Schoeman, J. N. Smith, L. E. Trigg, and C. B. Embling. "The Effects of Ship Noise on Marine Mammals—A Review". *Front. Mar. Sci.* 6 (2019), 606.
6. J. M. Fields. "Effect of Personal and Situational Variables on Noise Annoyance in Residential Areas". *J. Acoust. Soc. Am.* 93.5 (1993), 2753–2763.
7. C. R. Benítez-Barrera, N. Ramirez-Esparza, A. García-Sierra, and E. Skoe. "Cultural Differences in Auditory Ecology". *JASA Express Letters* 3.8 (2023), 083601.
8. N. Ramírez-Esparza, S. Jiang, A. García-Sierra, E. Skoe, and C. R. Benítez-Barrera. "Effects of Cultural Dynamics on Everyday Acoustic Environments". *J. Acoust. Soc. Am.* 156.3 (2024), 1942–1951.
9. J.-C. Valière, B. Palazzo-Bertholon, J.-D. Polack, and P. Carvalho. "Acoustic Pots in Ancient and Medieval Buildings: Literary Analysis of Ancient Texts and Comparison with Recent Observations in French Churches". *Acta Acust.* 99.1 (2013), 70–81.
10. N. F. Declercq and C. S. A. Dekeyser. "Acoustic Diffraction Effects at the Hellenistic Amphitheater of Epidaurus: Seat Rows Responsible for the Marvelous Acoustics". *J. Acoust. Soc. Am.* 121.4 (2007), 2011–2022.
11. J.-F. Allard and N. Atalla. *Propagation of Sound in Porous Media: Modelling Sound Absorbing Materials*. 2nd ed. Hoboken, NJ: Wiley, 2009.
12. M. Bruneau. *Fundamentals of Acoustics*. London: ISTE, 2006.
13. J. Auriault, C. Geindreau, P. Royer, J. Bloch, C. Boutin, and J. Lewandowska. "Poromechanics II". *Rotterdam, Brookfield* (2002).
14. P. Liu and G.-F. Chen. *Porous materials: processing and applications*. Elsevier, 2014.
15. T. Bourbié, O. Coussy, and B. Zinszner. *Acoustique des milieux poreux*. 27. éditions Technip, 1986.
16. O. Coussy. *Mechanics and physics of porous solids*. John Wiley & Sons, 2011.
17. D. Lafarge. "Propagation du son dans les matériaux poreux à structure rigide saturés par un fluide viscothermique: Définition de paramètres géométriques, analogie électromagnétique, temps de relaxation". PhD thesis. Le Mans, 1993.
18. D. Lafarge, J. F. Allard, B. Brouard, C. Verhaegen, and W. Lauriks. "Characteristic dimensions and prediction at high frequencies of the surface impedance of porous layers". *The Journal of the Acoustical Society of America* 93.5 (1993), 2474–2478.
19. D. Lafarge, P. Lemarinier, J. F. Allard, and V. Tarnow. "Dynamic Compressibility of Air in Porous Structures at Audible Frequencies". *J. Acoust. Soc. Am.* 102.4 (1997), 1995–2006.
20. A. Duclos, D. Lafarge, V. Pagneux, and A. Cortis. "Multiple scattering and visco-thermal effects". *The Journal of the Acoustical Society of America* 120.5 (2006), 3282–3282.
21. B. Brouard, D. Lafarge, and J.-F. Allard. "A general method of modelling sound propagation in layered media". *Journal of Sound and Vibration* 183.1 (1995), 129–142.
22. V. G. Veselago. "THE ELECTRODYNAMICS OF SUBSTANCES WITH SIMULTANEOUSLY NEGATIVE VALUES OF ϵ AND μ ". *Sov. Phys. Usp.* 10.4 (1968), 509–514.
23. P. Russell. "Photonic Crystal Fibers". *Science* 299.5605 (2003), 358–362.
24. E. Yablonovitch. "Inhibited Spontaneous Emission in Solid-State Physics and Electronics". *Phys. Rev. Lett.* 58.20 (1987), 2059–2062.
25. M. Sigalas and E. Economou. "Elastic and Acoustic Wave Band Structure". *J. Sound Vib.* 158.2 (1992), 377–382.
26. R. Martínez-Sala, J. Sancho, J. V. Sánchez, V. Gómez, J. Llinares, and F. Meseguer. "Sound Attenuation by Sculpture". *Nature* 378.6554 (1995), 241–241.
27. N. Jiménez, O. Umnova, and J.-P. Groby, eds. *Acoustic Waves in Periodic Structures, Metamaterials, and Porous Media: From Fundamentals to Industrial Applications*. Vol. 143. Topics in Applied Physics. Cham: Springer International Publishing, 2021.
28. X. Zhou, X. Wang, and F. Xin. "Ultrathin Acoustic Metamaterial as Super Absorber for Broadband Low-Frequency Underwater Sound". *Sci Rep* 13.1 (2023), 7983.
29. I. Prasetyo, I. Sihar, and A. S. Sudarsono. "Realization of a Thin and Broadband Microperforated Panel (MPP) Sound Absorber". *Applied Acoustics* 183 (2021), 108295.
30. V. Romero-García and A. Hladky-Hennion. *Fundamentals and Applications of Acoustic Metamaterials: From Seismic to Radio Frequency*. ISTE Wiley, 2019.

31. P. A. Deymier. *Acoustic metamaterials and phononic crystals*. Vol. 173. Springer Science & Business Media, 2013.
32. N. Jiménez, O. Umnova, and J.-P. Groby. “Acoustic Waves in Periodic Structures, Metamaterials, and Porous Media” (2021).
33. Z. Liu et al. “Locally Resonant Sonic Materials” ().
34. N. Jiménez, V. Romero-García, A. Cebrecos, R. Picó, V. J. Sánchez-Morcillo, and L. M. Garcia-Raffi. “Broadband Quasi Perfect Absorption Using Chirped Multi-Layer Porous Materials”. *AIP Advances* 6.12 (2016), 121605.
35. Y. Li and B. M. Assouar. “Acoustic Metasurface-Based Perfect Absorber with Deep Subwavelength Thickness”. *Appl. Phys. Lett.* 108.6 (2016), 063502.
36. K. Y. Bliokh, Y. P. Bliokh, V. Freilikher, S. Savel'ev, and F. Nori. “Colloquium: Unusual resonators: Plasmonics, metamaterials, and random media”. *Reviews of Modern Physics* 80.4 (2008), 1201.
37. J.-P. Groby, R. Pommier, and Y. Aurégan. “Use of slow sound to design perfect and broadband passive sound absorbing materials”. *The Journal of the Acoustical Society of America* 139.4 (2016), 1660–1671.
38. N. Jiménez, V. Romero-García, V. Pagneux, and J.-P. Groby. “Rainbow-trapping absorbers: Broadband, perfect and asymmetric sound absorption by subwavelength panels for transmission problems”. *Scientific reports* 7.1 (2017), 1–12.
39. V. Romero-García, G. Theocharis, O. Richoux, and V. Pagneux. “Use of complex frequency plane to design broadband and subwavelength absorbers”. *The Journal of the Acoustical Society of America* 139.6 (2016), 3395–3403.
40. V. Romero-García, G. Theocharis, O. Richoux, A. Merkel, V. Tournat, and V. Pagneux. “Perfect and broadband acoustic absorption by critically coupled sub-wavelength resonators”. *Sci. Rep.* 6 (2016), 19519EP.
41. J. Guo, Y. Fang, R. Qu, Q. Liu, and X. Zhang. “An Extra-Broadband Compact Sound-Absorbing Structure Composing of Double-Layer Resonator with Multiple Perforations”. *J. Acoust. Soc. Am.* 150.2 (2021), 1370–1380.
42. M. Yang and P. Sheng. “Sound Absorption Structures: From Porous Media to Acoustic Metamaterials”. *Annual Review of Materials Research* 47.1 (2017), 83–114.
43. Y. Aurégan, M. Farooqui, and J.-P. Groby. “Low frequency sound attenuation in a flow duct using a thin slow sound material”. *The Journal of the Acoustical Society of America* 139.5 (2016), EL149–EL153.
44. J. Yang, J. S. Lee, H. R. Lee, Y. J. Kang, and Y. Y. Kim. “Slow-wave metamaterial open panels for efficient reduction of low-frequency sound transmission”. *Applied Physics Letters* 112.9 (2018), 091901.
45. J. Leng, F. Gautier, A. Pelat, R. Pico, J.-P. Groby, and V. Romero-García. “Limits of flexural wave absorption by open lossy resonators: reflection and transmission problems”. *New Journal of Physics* 21.5 (2019), 053003.
46. A. Lardeau, J.-P. Groby, and V. Romero-García. “Broadband transmission loss using the overlap of resonances in 3D sonic crystals”. *Crystals* 6.5 (2016), 51.
47. J. Yang, J. S. Lee, and Y. Y. Kim. “Metaporous Layer to Overcome the Thickness Constraint for Broadband Sound Absorption”. *J. Appl. Phys.* 117.17 (2015), 174903.
48. X. Wang, X. Luo, H. Zhao, and Z. Huang. “Acoustic perfect absorption and broadband insulation achieved by double-zero metamaterials”. *Applied Physics Letters* 112.2 (2018), 021901.
49. V. Romero-García, G. Theocharis, O. Richoux, and V. Pagneux. “Use of complex frequency plane to design broadband and subwavelength absorbers”. *The Journal of the Acoustical Society of America* 139.6 (2016), 3395–3403.
50. V. Romero-García et al. “Perfect absorption in mirror-symmetric acoustic metascreens”. *Physical Review Applied* 14.5 (2020), 054055.
51. H. Long, Y. Cheng, and X. Liu. “Asymmetric absorber with multiband and broadband for low-frequency sound”. *Applied Physics Letters* 111.14 (2017), 143502.
52. H. Long et al. “Tunable and broadband asymmetric sound absorptions with coupling of acoustic bright and dark modes”. *Journal of Sound and Vibration* 479 (2020), 115371.
53. M. Yang, G. Ma, Z. Yang, and P. Sheng. “Coupled Membranes with Doubly Negative Mass Density and Bulk Modulus”. *Phys. Rev. Lett.* 110.13 (2013), 134301.
54. Z. Yang, J. Mei, M. Yang, N. H. Chan, and P. Sheng. “Membrane-Type Acoustic Metamaterial with Negative Dynamic Mass”. *Phys. Rev. Lett.* 101.20 (2008), 204301.
55. A. Merkel, G. Theocharis, O. Richoux, V. Romero-García, and V. Pagneux. “Control of acoustic absorption in one-dimensional scattering by resonant scatterers”. *Applied Physics Letters* 107.24 (2015), 244102.
56. N. Jiménez, V. Romero-García, V. Pagneux, and J.-P. Groby. “Quasiperfect absorption by subwavelength acoustic panels in transmission using accumulation of resonances due to slow sound”. *Phys. Rev. B* 95 (1 2017), 014205.
57. J. R. Piper and S. Fan. “Total absorption in a graphene monolayer in the optical regime by critical coupling with a photonic crystal guided resonance”. *ACS Photonics* 1.4 (2014), 347–353.
58. C. Fu, X. Zhang, M. Yang, S. Xiao, and Z. Yang. “Hybrid membrane resonators for multiple frequency asymmetric absorption and reflection in large waveguide”. *Applied Physics Letters* 110.2 (2017), 021901.
59. J. Boulvert et al. “Optimally Graded Porous Material for Broadband Perfect Absorption of Sound”. *J. Appl. Phys.* 126.17 (2019), 175101.
60. J. Boulvert et al. “Acoustic modeling of micro-lattices obtained by additive manufacturing”. *Applied Acoustics* 164 (2020), 107244.
61. T. Cavalieri et al. “Acoustic wave propagation in effective graded fully anisotropic fluid layers”. *The Journal of the Acoustical Society of America* 146.5 (2019), 3400–3408.
62. T. Cavalieri et al. “Graded and Anisotropic Porous Materials for Broadband and Angular Maximal Acoustic Absorption”. *Materials* 13.20 (2020), 4605.
63. T. Cavalieri et al. “Rapid Additive Manufacturing of Optimized Anisotropic Metaporous Surfaces for Broadband Absorption”. *J. Appl. Phys.* 129.11 (2021), 115102.

64. A. Terroir, L. Schwan, T. Cavaliere, V. Romero-García, G. Gabard, and J.-P. Groby. “General method to retrieve all effective acoustic properties of fully-anisotropic fluid materials in three dimensional space”. *Journal of Applied Physics* 125.2 (2019), 025114.
65. J. Boulvert et al. “Folded metaporous material for sub-wavelength and broadband perfect sound absorption”. *Applied Physics Letters* 117.25 (2020), 251902.
66. J. Yang, J. S. Lee, and Y. Y. Kim. “Metaporous Layer to Overcome the Thickness Constraint for Broadband Sound Absorption”. *Journal of Applied Physics* 117.17 (2015).
67. X. Liu, M. Duan, M. Liu, F. Xin, and C. Zhang. “Acoustic Labyrinthine Porous Metamaterials for Subwavelength Low-Frequency Sound Absorption”. *Journal of Applied Physics* 129.19 (2021), 195103.
68. S. Kuznetsova, S. Deleplanque, F. Allein, B. Dubus, and M. Miniaci. “Hierarchical Meta-Porous Materials as Sound Absorbers”. *Proc. R. Soc. A* 480.2301 (2024), 20230831.
69. J.-P. Groby, O. Dazel, A. Duclos, L. Boeckx, and W. Lauriks. “Quasi-total absorption peak by use of a backed rigid frame porous layer with circular periodic inclusions embedded”. *arXiv preprint arXiv:1007.3177* (2010).
70. J.-P. Groby, O. Dazel, A. Duclos, L. Boeckx, and L. Kelders. “Enhancing the Absorption Coefficient of a Backed Rigid Frame Porous Layer by Embedding Circular Periodic Inclusions”. *J. Acoust. Soc. Am.* 130.6 (2011), 3771–3780.
71. C. Lagarrigue, J. P. Groby, V. Tournat, O. Dazel, and O. Umnova. “Absorption of Sound by Porous Layers with Embedded Periodic Arrays of Resonant Inclusions”. *J. Acoust. Soc. Am.* 134.6 (2013), 4670–4680.
72. J.-P. Groby, C. Lagarrigue, B. Brouard, O. Dazel, V. Tournat, and B. Nennig. “Using simple shape three-dimensional rigid inclusions to enhance porous layer absorption”. *The Journal of the Acoustical Society of America* 136.3 (2014), 1139–1148.
73. J.-P. Groby, A. Wirgin, and E. Ogam. “Acoustic response of a periodic distribution of macroscopic inclusions within a rigid frame porous plate”. *Waves in Random and Complex Media* 18.3 (2008), 409–433.
74. L. De Ryck et al. “Acoustic wave propagation and internal fields in rigid frame macroscopically inhomogeneous porous media”. *Journal of Applied Physics* 102.2 (2007), 024910.
75. J.-P. Groby, C. Lagarrigue, B. Brouard, O. Dazel, V. Tournat, and B. Nennig. “Enhancing the Absorption Properties of Acoustic Porous Plates by Periodically Embedding Helmholtz Resonators”. *J. Acoust. Soc. Am.* 137.1 (2015), 273–280.
76. Y. Zhou, D. Li, Y. Li, and T. Hao. “Perfect Acoustic Absorption by Subwavelength Metaporous Composite”. *Applied Physics Letters* 115.9 (2019), 093503.
77. L. Xiong, B. Nennig, Y. Aurégan, and W. Bi. “Sound Attenuation Optimization Using Metaporous Materials Tuned on Exceptional Points”. *J. Acoust. Soc. Am.* 142.4 (2017), 2288–2297.
78. T. Weisser et al. “Acoustic Behavior of a Rigidly Backed Poroelastic Layer with Periodic Resonant Inclusions by a Multiple Scattering Approach”. *J. Acoust. Soc. Am.* 139.2 (2016), 617–629.
79. M. Gaborit, L. Schwan, O. Dazel, J.-P. Groby, T. Weisser, and P. Göransson. “Coupling FEM, Bloch Waves and TMM in Meta Poroelastic Laminates”. *Acta Acust.* 104.2 (2018), 220–227.
80. C. Prada, D. Clorennec, and D. Royer. “Local Vibration of an Elastic Plate and Zero-Group Velocity Lamb Modes”. *The Journal of the Acoustical Society of America* 124.1 (2008), 203–212.
81. L. Boeckx, P. Leclaire, P. Khurana, C. Glorieux, W. Lauriks, and J. F. Allard. “Guided Elastic Waves in Porous Materials Saturated by Air under Lamb Conditions”. *J. Appl. Phys.* 97.9 (2005), 094911.
82. L. Boeckx, P. Leclaire, P. Khurana, C. Glorieux, W. Lauriks, and J. F. Allard. “Investigation of the Phase Velocities of Guided Acoustic Waves in Soft Porous Layers”. *J. Acoust. Soc. Am.* 117.2 (2005), 545–554.
83. D. A. Kiefer, M. Ponschab, and S. J. Rupitsch. “From Lamb Waves to Quasi-Guided Waves: On the Wave Field and Radiation of Elastic and Viscoelastic Plates”. *Unpublished* (2020).
84. M. Delany and E. Bazley. “Acoustical Properties of Fibrous Absorbent Materials”. *Applied Acoustics* 3.2 (1970), 105–116.
85. Y. Miki. “Acoustical Properties of Porous Materials. Modifications of Delany-Bazley Models”. *J. Acoust. Soc. Jpn. (E), J Acoust Soc Jpn E* 11.1 (1990), 19–24.
86. M. A. Biot. “Theory of Stress-Strain Relations in Anisotropic Viscoelasticity and Relaxation Phenomena”. *J. Appl. Phys.* 25.11 (1954), 1385–1391.
87. M. A. Biot. “Theory of Propagation of Elastic Waves in a Fluid-Saturated Porous Solid. II. Higher Frequency Range”. *en. The Journal of the Acoustical Society of America* 28.2 (1956), 179–191.
88. M. A. Biot. “Theory of Propagation of Elastic Waves in a Fluid-Saturated Porous Solid. I. Low-Frequency Range”. *J. Acoust. Soc. Am.* 28.2 (1956), 168–178.
89. M. A. Biot. “Theory of Propagation of Elastic Waves in a Fluid-Saturated Porous Solid. II. Higher Frequency Range”. *J. Acoust. Soc. Am.* 28.2 (1956), 179–191.
90. N. Atalla, R. Panneton, and P. Debergue. “A Mixed Displacement-Pressure Formulation for Poroelastic Materials”. *J. Acoust. Soc. Am.* 104.3 (1998), 1444–1452.
91. O. Doutres, N. Atalla, and H. Osman. “Transfer Matrix Modeling and Experimental Validation of Cellular Porous Material with Resonant Inclusions”. *J. Acoust. Soc. Am.* 137.6 (2015), 3502–3513.
92. L. N. Trefethen. *Spectral Methods in MATLAB*. Society for Industrial and Applied Mathematics, 2000.
93. P. Kowalczyk and W. Marynowski. “Efficient Complex Root Tracing Algorithm for Propagation and Radiation Problems”. *IEEE Trans. Antennas Propagat.* 65.5 (2017), 2540–2546.
94. P. S. Dobbelday. “Application of a New Complex Root-Finding Technique to the Dispersion Relations for Elastic Waves in a Fluid-Loaded Plate”. *SIAM Journal on Applied Mathematics* 43.5 (1983), 1127–1139.
95. D. E. Muller. “A Method for Solving Algebraic Equations Using an Automatic Computer”. *Math. Comp.* 10.56 (1956), 208.

96. B. Pavlakovic, M. Lowe, D. Alleyne, and P. Cawley. "Disperse: A General Purpose Program for Creating Dispersion Curves". In: *Review of Progress in Quantitative Nondestructive Evaluation*. Ed. by D. O. Thompson and D. E. Chimenti. Boston, MA: Springer US, 1997, 185–192.
97. A. M. A. Huber and M. G. R. Sause. "Classification of Solutions for Guided Waves in Anisotropic Composites with Large Numbers of Layers". *J. Acoust. Soc. Am.* 144.6 (2018), 3236–3251.
98. A. M. A. Huber. "Classification of Solutions for Guided Waves in Fluid-Loaded Viscoelastic Composites with Large Numbers of Layers". *J. Acoust. Soc. Am.* 154.2 (2023), 1073–1094.
99. F. Treysède, K. Nguyen, A.-S. Bonnet-BenDhia, and C. Hazard. "Finite Element Computation of Trapped and Leaky Elastic Waves in Open Stratified Waveguides". *Wave Motion* 51.7 (2014), 1093–1107.
100. I. Bartoli, A. Marzani, F. Lanza Di Scalea, and E. Viola. "Modeling Wave Propagation in Damped Waveguides of Arbitrary Cross-Section". *J. Sound Vib.* 295.3-5 (2006), 685–707.
101. A. Marzani, E. Viola, I. Bartoli, F. Lanza Di Scalea, and P. Rizzo. "A Semi-Analytical Finite Element Formulation for Modeling Stress Wave Propagation in Axisymmetric Damped Waveguides". *J. Sound Vib.* 318.3 (2008), 488–505.
102. J. L. Rose. *Ultrasonic Guided Waves in Solid Media*. 1st ed. Cambridge University Press, 2014.
103. C. Canuto, ed. *Spectral Methods: Fundamentals in Single Domains; with 19 Tables*. Scientific Computation. Berlin Heidelberg New York: Springer, 2006.
104. M. Lowe. "Matrix Techniques for Modeling Ultrasonic Waves in Multilayered Media". *IEEE Trans. Ultrason., Ferroelect., Freq. Contr.* 42.4 (1995), 525–542.
105. N. A. Haskell. "The Dispersion of Surface Waves on Multilayered Media*". *Bulletin of the Seismological Society of America* 43.1 (1953), 17–34.
106. V. Pagneux and A. Maurel. "Determination of Lamb Mode Eigenvalues". *J. Acoust. Soc. Am.* 110.3 (2001), 1307–1314.
107. M. Gallezot, F. Treysède, and L. Laguerre. "Contribution of Leaky Modes in the Modal Analysis of Unbounded Problems with Perfectly Matched Layers". *J. Acoust. Soc. Am.* 141.1 (2017), EL16–EL21.
108. E. Georgiades, M. J. S. Lowe, and R. V. Craster. "Leaky Wave Characterisation Using Spectral Methods". *J. Acoust. Soc. Am.* 152.3 (2022), 1487–1497.
109. D. A. Kiefer, M. Ponschab, S. J. Rupitsch, and M. Mayle. "Calculating the Full Leaky Lamb Wave Spectrum with Exact Fluid Interaction". *J. Acoust. Soc. Am.* 145.6 (2019), 3341–3350.
110. M. A. Biot. "Mechanics of Deformation and Acoustic Propagation in Porous Media". *J. Appl. Phys.* 33.4 (1962), 1482–1498.
111. M. A. Biot. "Generalized Theory of Acoustic Propagation in Porous Dissipative Media". *J. Acoust. Soc. Am.* 34.9A (1962), 1254–1264.
112. D. L. Johnson, J. Koplik, and R. Dashen. "Theory of Dynamic Permeability and Tortuosity in Fluid-Saturated Porous Media". *J. Fluid Mech.* 176.-1 (1987), 379.
113. Y. Champoux and J.-F. Allard. "Dynamic Tortuosity and Bulk Modulus in Air-saturated Porous Media". *J. Appl. Phys.* 70.4 (1991), 1975–1979.
114. G. Belloncle, H. Franklin, F. Luppé, and J. Conoir. "Normal Modes of a Poroelastic Plate and Their Relation to the Reflection and Transmission Coefficients". *Ultrasonics* 41.3 (2003), 207–216.
115. J. F. Allard, M. Henry, and V. Garetton. "Pseudo-Surface Waves above Thin Porous Layers (L)". *J. Acoust. Soc. Am.* 116.3 (2004), 1345–1347.
116. F. Seyfaddini, H. Nguyen-Xuan, and V.-H. Nguyen. "Semi-Analytical IGA-based Computation of Wave Dispersion in Fluid-Coupled Anisotropic Poroelastic Plates". *Int. J. Mech. Sci.* 212 (2021), 106830.
117. A. T. I. Adamou and R. V. Craster. "Spectral Methods for Modelling Guided Waves in Elastic Media". *J. Acoust. Soc. Am.* 116.3 (2004), 1524–1535.
118. F. Karpfinger, B. Gurevich, and A. Bakulin. "Modeling of Wave Dispersion along Cylindrical Structures Using the Spectral Method". *J. Acoust. Soc. Am.* 124.2 (2008), 859–865.
119. F. H. Quintanilla, M. J. S. Lowe, and R. V. Craster. "Modeling Guided Elastic Waves in Generally Anisotropic Media Using a Spectral Collocation Method". *J. Acoust. Soc. Am.* 137.3 (2015), 1180–1194.
120. A. Hood. "Localizing the Eigenvalues of Matrix-Valued Functions: Analysis and Applications". PhD thesis. Cornell University Library, 2017.
121. T. Betcke, N. J. Higham, V. Mehrmann, C. Schröder, and F. Tisseur. "NLEVP: A Collection of Nonlinear Eigenvalue Problems". *ACM Trans. Math. Softw.* 39.2 (2013), 1–28.
122. I. Gohberg, P. Lancaster, and L. Rodman. *Matrix Polynomials*. Society for Industrial and Applied Mathematics, 2009.
123. H. Gravenkamp, B. Plestenjak, D. A. Kiefer, and E. Jarlebring. *Computation of Leaky Waves in Layered Structures Coupled to Unbounded Media by Exploiting Multiparameter Eigenvalue Problems*. 2024.
124. J. A. Weideman and S. C. Reddy. "A MATLAB Differentiation Matrix Suite". *ACM Trans. Math. Softw.* 26.4 (2000), 465–519.
125. R. Baltensperger and M. R. Trummer. "Spectral Differencing with a Twist". *SIAM J. Sci. Comput.* 24.5 (2003), 1465–1487.
126. G. Strang. *Introduction to Linear Algebra*. SIAM, 2022.
127. J. D. Achenbach, H. A. Lauwerier, and W. T. Koiter. *Wave Propagation in Elastic Solids: North-Holland Series in Applied Mathematics and Mechanics*. Amsterdam: Elsevier Science, 2014.
128. P. Debergue, R. Panneton, and N. Atalla. "Boundary Conditions for the Weak Formulation of the Mixed (u,p) Poroelasticity Problem". *J. Acoust. Soc. Am.* 106.5 (1999), 2383–2390.
129. L. Brillouin. *Wave Propagation in Periodic Structures*. MGH, 1946.
130. A. Bernard, M. J. S. Lowe, and M. Deschamps. "Guided Waves Energy Velocity in Absorbing and Non-Absorbing Plates". *J. Acoust. Soc. Am.* 110.1 (2001), 186–196.
131. F. Simonetti and M. J. S. Lowe. "On the Meaning of Lamb Mode Nonpropagating Branches". *J. Acoust. Soc. Am.* 118.1 (2005), 186–192.

132. M. Castaings and B. Hosten. "Guided Waves Propagating in Sandwich Structures Made of Anisotropic, Viscoelastic, Composite Materials". *J. Acoust. Soc. Am.* 113.5 (2003), 2622–2634.
133. J. M. Carcione. "Wave Propagation in Anisotropic, Saturated Porous Media: Plane-wave Theory and Numerical Simulation". *J. Acoust. Soc. Am.* 99.5 (1996), 2655–2666.
134. A. Geslain et al. "Spatial Laplace Transform for Complex Wavenumber Recovery and Its Application to the Analysis of Attenuation in Acoustic Systems". *J. Appl. Phys.* 120.13 (2016), 135107.
135. O. Dazel, F. Sgard, F.-X. Becot, and N. Atalla. "Expressions of Dissipated Powers and Stored Energies in Poroelastic Media Modeled by (u,U) and (u,P) Formulations". *J. Acoust. Soc. Am.* 123.4 (2008).
136. D. Royer and T. Valier-Brasier. *Ondes élastiques dans les solides*. Collection Ondes. London: ISTE éditions, 2021.
137. M. V. Predoi, M. Castaings, B. Hosten, and C. Bacon. "Wave Propagation along Transversely Periodic Structures". *J. Acoust. Soc. Am.* 121.4 (2007), 1935–1944.
138. M. Castaings and M. Lowe. "Finite Element Model for Waves Guided along Solid Systems of Arbitrary Section Coupled to Infinite Solid Media". *J. Acoust. Soc. Am.* 123.2 (2008), 696–708.
139. P. Zuo and Z. Fan. "SAFE-PML Approach for Modal Study of Waveguides with Arbitrary Cross Sections Immersed in Inviscid Fluid". *J. Sound Vib.* 406 (2017), 181–196.
140. M. Mazzotti, I. Bartoli, A. Marzani, and E. Viola. "A Coupled SAFE-2.5D BEM Approach for the Dispersion Analysis of Damped Leaky Guided Waves in Embedded Waveguides of Arbitrary Cross-Section". *Ultrasonics* 53.7 (2013), 1227–1241.
141. M. Mazzotti, A. Marzani, and I. Bartoli. "Dispersion Analysis of Leaky Guided Waves in Fluid-Loaded Waveguides of Generic Shape". *Ultrasonics* 54.1 (2014), 408–418.
142. T. Hayashi and D. Inoue. "Calculation of Leaky Lamb Waves with a Semi-Analytical Finite Element Method". *Ultrasonics* 54.6 (2014), 1460–1469.
143. S. C. Chapra and R. P. Canale. *Numerical Methods for Engineers*. 6th ed. Boston: McGraw-Hill Higher Education, 2010.